

CSE4225 Digital Image Processing

Chapter 4: Image Enhancement in the Frequency Domain

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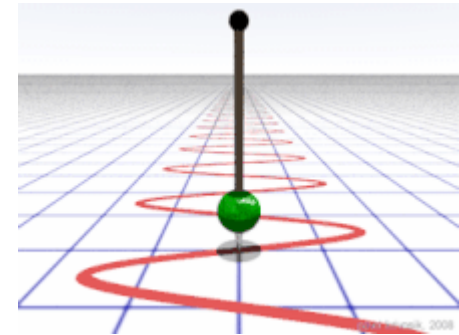
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Enhancement: To make greater <as in value, desirability, or attractiveness>

Frequency: The number of times that a periodic function repeats the same sequence of values during a unit variation of the independent variable.

Webster's New Collegiate Dictionary



A pendulum making 25 complete oscillations in 60 s,
a frequency of 0.42 Hertz

Fourier Transform

- A complicated signal can be broken down into simple waves
- This break down, and how much each wave is needed is the Fourier Transform



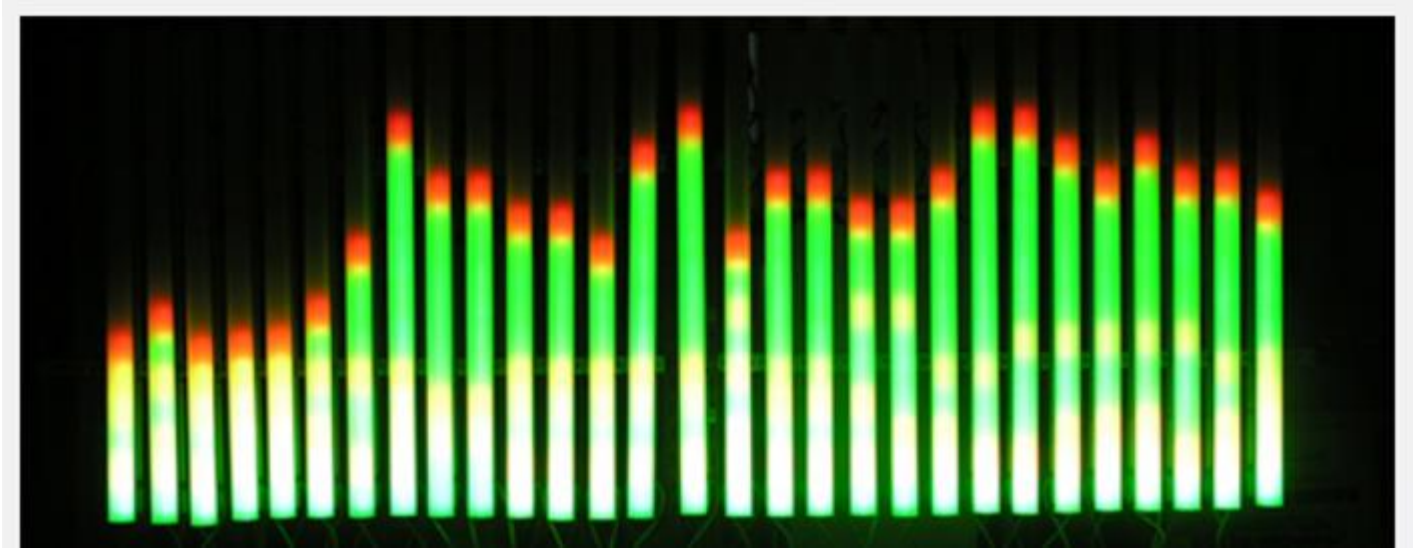
An Example

- The layout of the keys of a piano (bottom) are like the Fourier transform of the sound the piano makes (top)



An Example (Cont.)

- An example of a Fourier transform as seen on the front of a sound system



Jean Baptiste Joseph Fourier

- Fourier was born in Auxerre, France in 1768
- Most famous for his work “La Théorie Analitique de la Chaleur” published in 1822
- Translated into English in 1878: “The Analytic Theory of Heat”
- Nobody paid much attention when the work was first published
- One of the most important mathematical theories in modern engineering

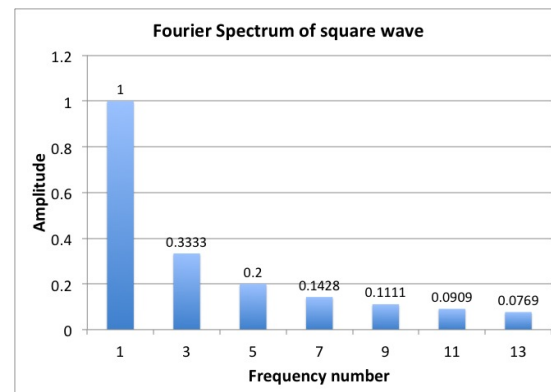
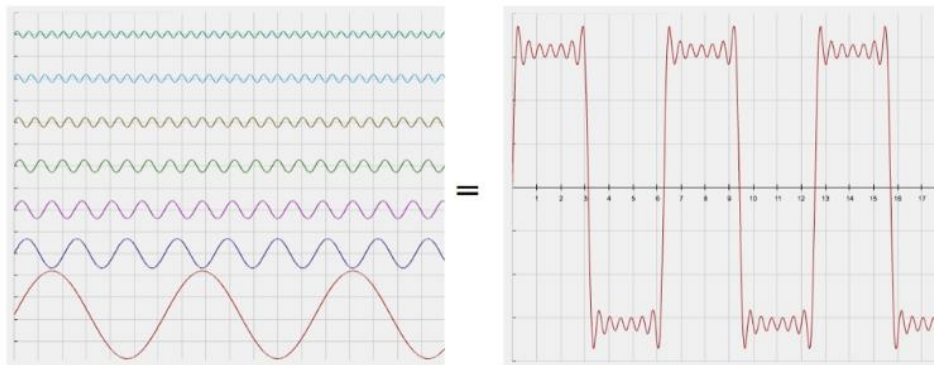


You know you are famous when
they put you on a mug!

The Big Idea

- Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient – a Fourier series
- For example: a square wave can be written as

$$s(t) = \sin(\omega t) + \frac{1}{3}\sin(3\omega t) + \frac{1}{5}\sin(5\omega t) + \frac{1}{7}\sin(7\omega t) + \dots$$

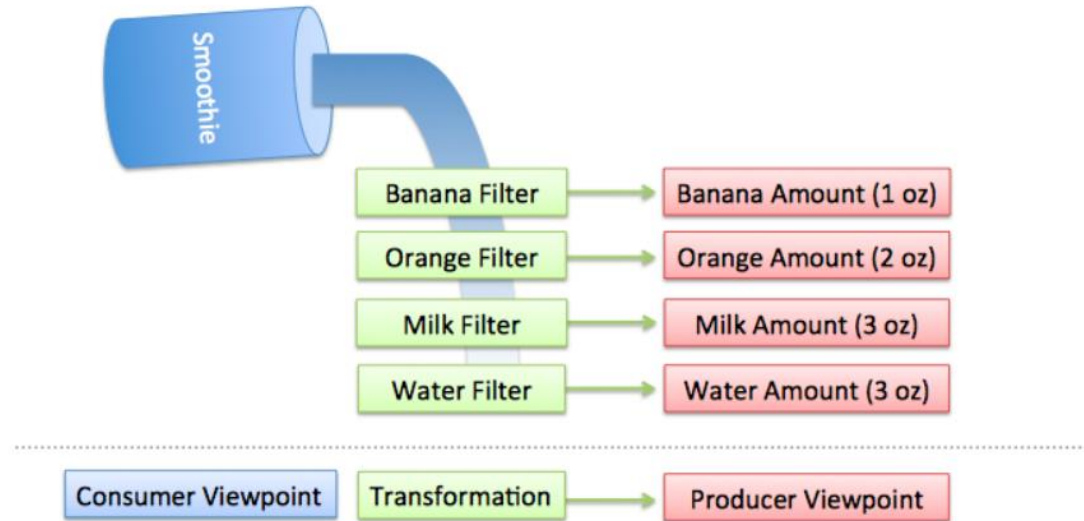


Fourier Transform (FT)

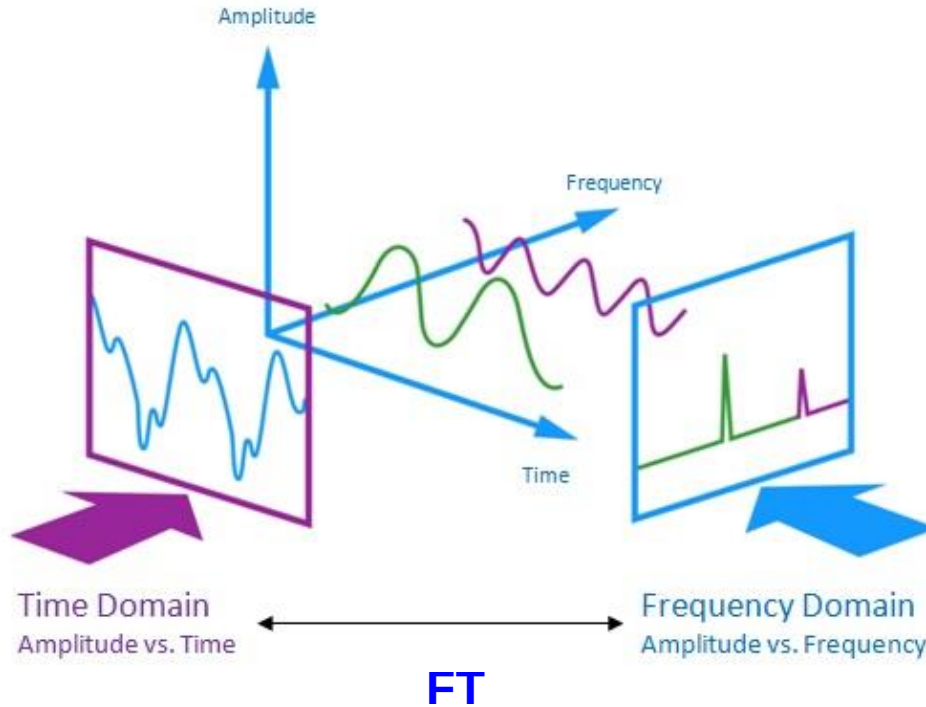
- Is an integral transform that re-expresses a function as a sum or integral of sinusoidal functions of different frequencies multiplied by some coefficients ("amplitudes")
- Takes a signal and express it in terms of the frequencies of the waves that make up that signal

Shootie to Recipe

- Fourier transforms (FT) finds the recipe for a signal



Time Domain Vs. Frequency Domain

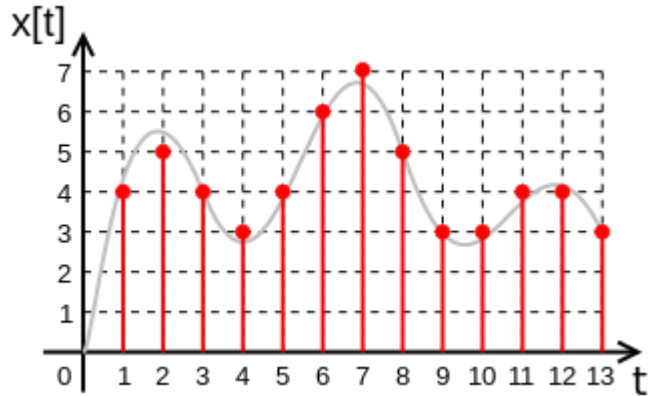


What if the function is not periodic??

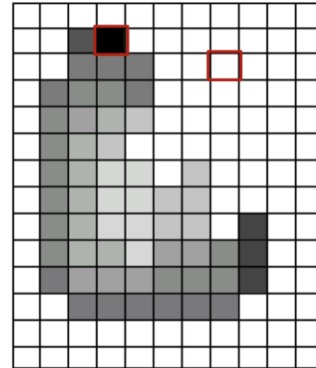
- Even the functions that are not periodic can be expressed as the integral of sines and/or cosines multiplied by a weighting functions using **Fourier Transform (FT)**

Discrete Fourier Transform (DFT)

- For a digitally sampled signal the discrete (or finite) Fourier transform is used



1 D Digital Signal



=

255	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255
255	255	20	0	255	255	255	255	255	255	255	255	255	255	255	255
255	255	75	75	255	255	255	255	255	255	255	255	255	255	255	255
255	75	95	95	75	255	255	255	255	255	255	255	255	255	255	255
255	96	127	145	175	255	255	255	255	255	255	255	255	255	255	255
255	127	145	175	175	175	255	255	255	255	255	255	255	255	255	255
255	127	145	200	200	175	175	95	255	255	255	255	255	255	255	255
255	127	145	200	200	175	175	95	47	255	255	255	255	255	255	255
255	127	145	145	175	127	127	95	47	255	255	255	255	255	255	255
255	74	127	127	127	95	95	95	47	255	255	255	255	255	255	255
255	255	74	74	74	74	74	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255

0 = black; 255 = white

2 D Digital Signal

1-D Discrete Fourier Transform (DFT)

Discrete function:

$$f(x), x = 0, 1, \dots, M-1$$

DFT of $f(x)$:

$$F(u), u = 0, 1, \dots, M-1$$

- Fourier Transform of $f(x)$:

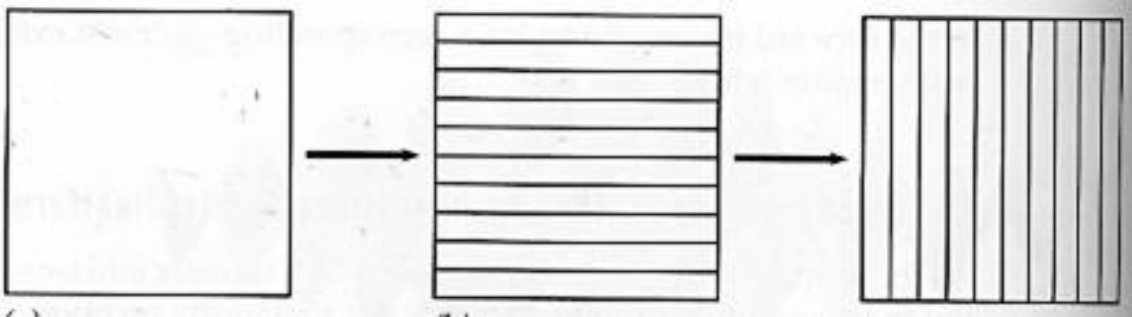
$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi \frac{u}{M}x}$$

- Inverse Transform (reconstruction):

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi \frac{u}{M}x}$$

Extend to 2-D DFT from 1-D

- 2-D DFT: 1-D DFT in horizontal then vertical



2D DFT

$x = 0, 1, 2 \dots M-1$ and $y = 0, 1, 2 \dots N-1$

DFT of $f(x, y)$:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$u = 0, 1, 2 \dots M-1$ and

$v = 0, 1, 2 \dots N-1$

Inverse of DFT

DFT is completely reversible.

The inverse DFT:

$u = 0, 1, 2 \dots M-1$ and $v = 0, 1, 2 \dots N-1$

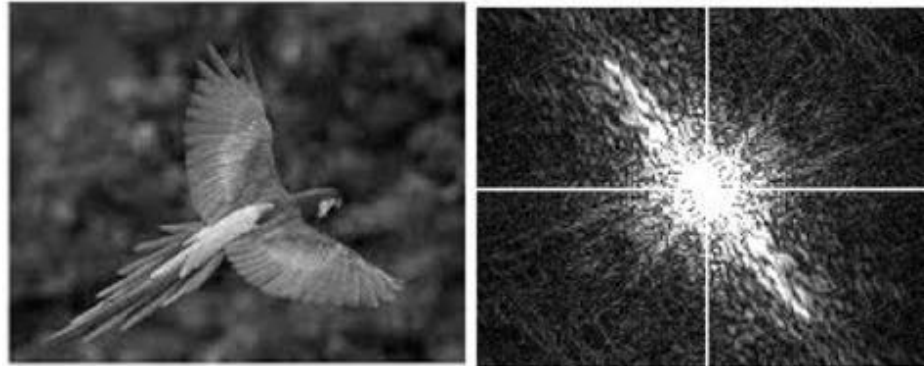
$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$x = 0, 1, 2 \dots M-1$ and

$y = 0, 1, 2 \dots N-1$

Fourier Transform of an Image

- Fourier Transform is used to convert images from the **spatial domain** into the **frequency domain** and vice-versa
- Applications: image analysis, image filtering, image reconstruction and image compression

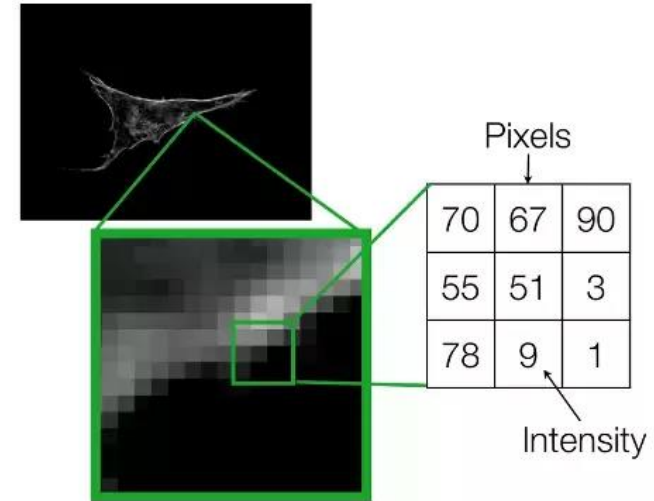


Spatial domain

Frequency domain

Spatial Domain

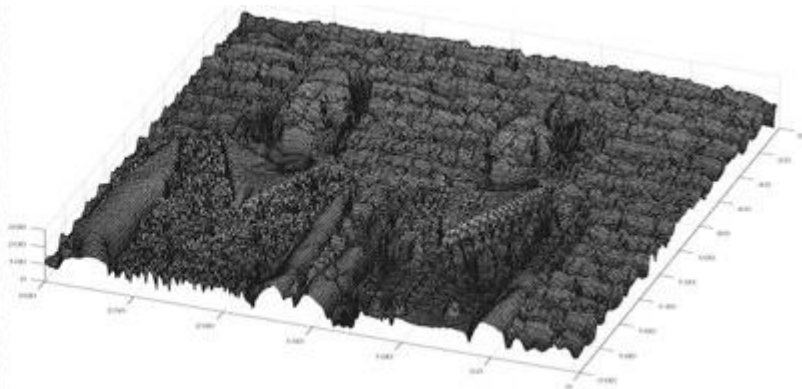
- Images are 2D functions $f(x,y)$ in spatial coordinates (x,y) in an image plane
 - (x,y) refers to the location of a pixel (smallest element in an image)
 - $f(x,y)$ refers to the pixel intensity
- Each function describes how colours or grey values (intensities, or brightness) vary in space
- **Spatial domain** refers to the image plane itself
- Methods in spatial domain are based on directly modifying the value of the pixels



DIP techniques we have used so far are in spatial domain

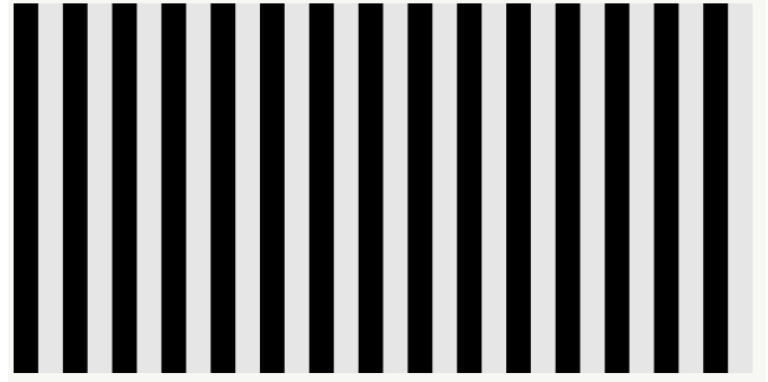
Frequency in an Image

- An image is a varying function
- It varies across the two-dimensional space of the image rather than varying in time
- Can also be expressed as a sum of sine waves those vary in two-dimensions, like ripples on a sheet



2D Spatial Frequency

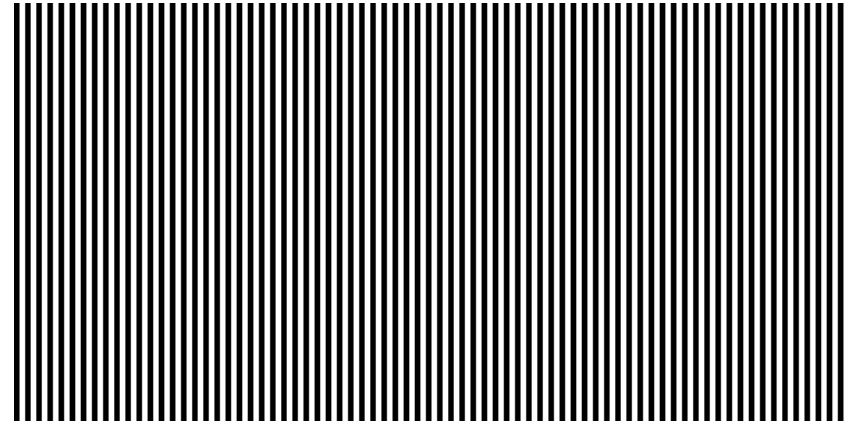
- Refers to the (inverse of the) periodicity with which the image intensity values change in the spatial domain image
- Do you see a pattern there ??
- There is a time period in the signal that is 40 pixels
 - After 40 pixels the signal repeats itself
- Spatial Frequency = $1/40 = 0.025$



600 x 300 px black and white image with
20 px wide stripes

2D Spatial Frequency

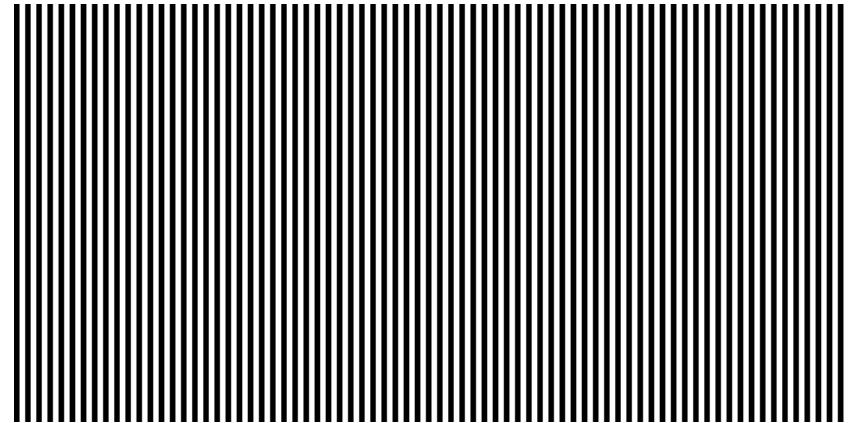
- What is the spatial frequency ??



600 x 300 px black and white image with 4
px wide stripes

2D Spatial Frequency

- What is the spatial frequency ??
- There is a time period in the signal that is 8 pixels
 - After 8 pixels the signal repeats itself
- Spatial Frequency = $1/8 = 0.125$



600 x 300 px black and white image with 4
px wide stripes

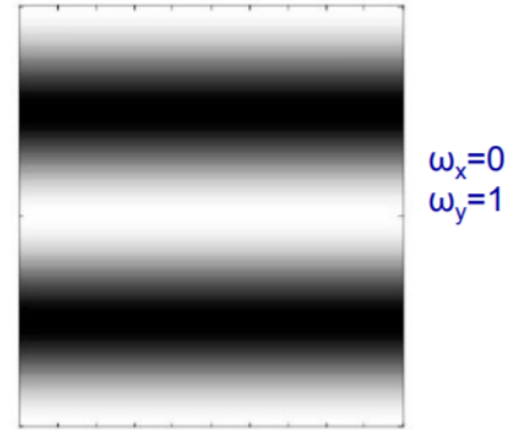
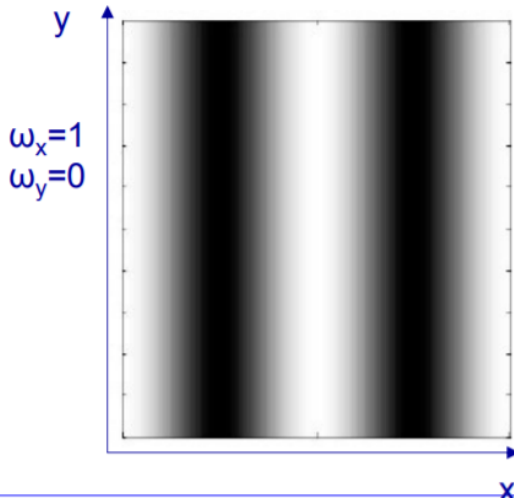
2D Spatial Frequency

- Maximum frequency which can be represented in the spatial domain is 0.5
- Happens when are two pixel wide stripe pairs (one white, one black)

$$f_{\max} = \frac{1}{2 \text{ pixels}}$$

2D Spatial Frequency

- 2D spatial frequencies characterize the image spatial changes in the horizontal (x) and vertical (y) directions
 - Smooth variations -> low frequencies
 - Sharp variations -> high frequencies

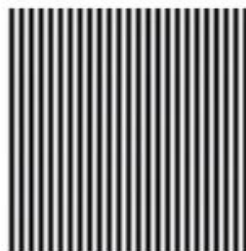


More 2D Spatial Frequencies



$$\cos\left(\frac{18\pi}{256}n\right)$$

$$u=0, v=\frac{18\pi}{256}$$



$$\cos\left(\frac{50\pi}{256}n\right)$$

$$u=0, v=\frac{50\pi}{256}$$



$$\cos\left(\frac{18\pi}{256}m\right)$$

$$u=\frac{18\pi}{256}, v=0$$



$$\cos\left(\frac{50\pi}{256}m\right)$$

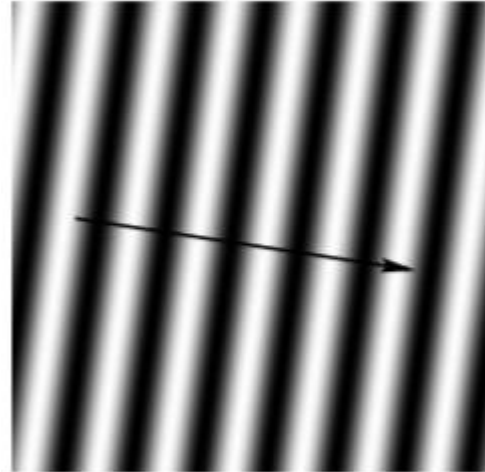
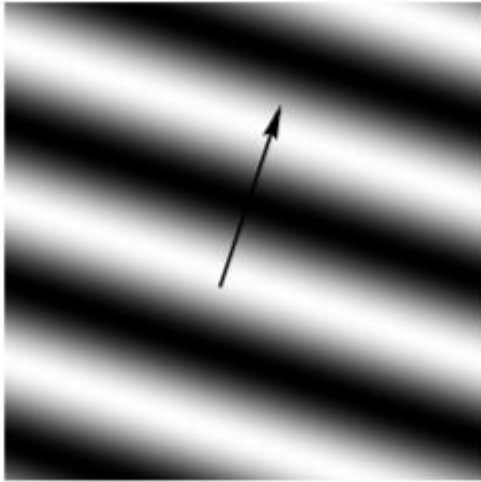
$$u=\frac{50\pi}{256}, v=0$$



$$\cos\left(\frac{50\pi}{256}n\right)\cos\left(\frac{18\pi}{256}m\right)$$

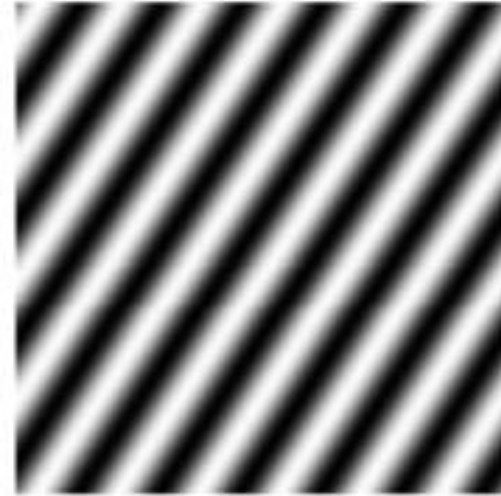
$$u=\frac{18\pi}{256}, v=\frac{50\pi}{256}$$

More 2D Spatial Frequencies



More 2D Spatial Frequencies

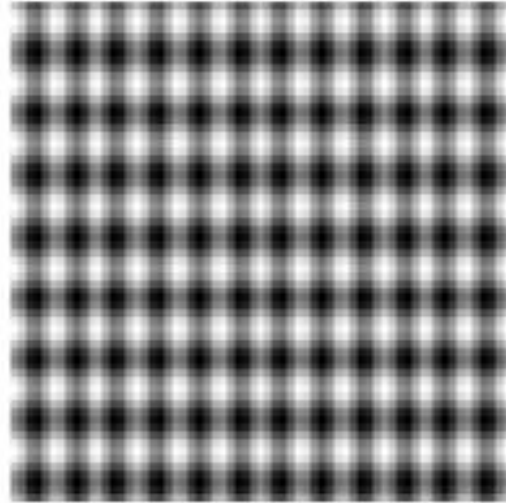
$$f(x, y) = \sin(2\pi x + 3\pi y)$$



```
[X,Y]=meshgrid(-2:1/16:2,-2:1/16:2);  
f=sin(2*pi*X+3*pi*Y);  
imagesc(f); colormap(gray)  
TrueSize, axis off;
```

More 2D Spatial Frequencies

$$f(x, y) = \sin 4\pi x + \cos 6\pi y$$



Frequency Domain of an Image

- FT decompose an image into its sine and cosine components
- The output of the transformation represents the image in the **Fourier or frequency domain (u, v)**
- While the input image is the **spatial domain (x, y)** equivalent
- In the Fourier domain image, each point represents a particular frequency contained in the spatial domain image
- Suppose that there is the value 20 at the point that represents the frequency 0.1 (or 1 period every 10 pixels)
- This means that in the corresponding spatial domain image I the intensity values vary from dark to light and back to dark over a distance of 10 pixels, and that the contrast between the lightest and darkest is 40 gray levels (2 times 20).

Calculating DFT

$$f(x, y) = R + jI$$

$$f(x, y)e^{-j\theta}$$

$$= (R + jI)(\cos\theta - j\sin\theta)$$

$$= R\cos\theta - jR\sin\theta + jI\cos\theta - j^2I\sin\theta$$

$$= R\cos\theta - jR\sin\theta + jI\cos\theta + I\sin\theta$$

$$= (R\cos\theta + I\sin\theta) + j(I\cos\theta - R\sin\theta)$$

$$= R\cos\theta + j(-R\sin\theta)$$

$$R = f(x, y), \quad \theta = 2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)$$

Calculating DFT (Cont.)

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} R \cos \theta + j \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (-R \sin \theta)$$

$$F(u, v) = a + jb$$

$$|F(u, v)| = (a^2 + b^2)^{1/2}$$



Magnitude

$$\phi(u, v) = \tan^{-1}\left(\frac{b}{a}\right)$$



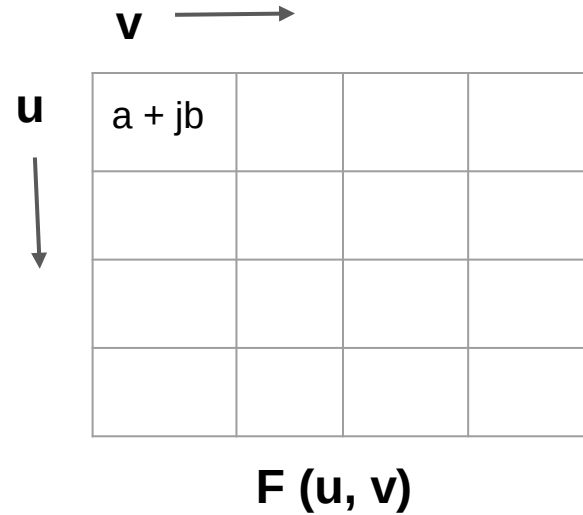
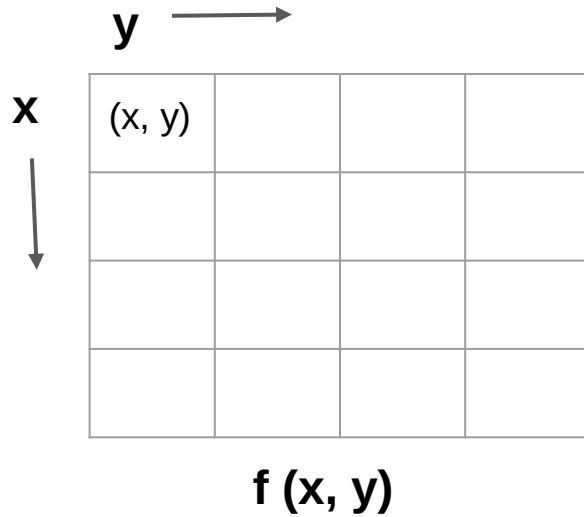
Phase

$$P(u, v) = |F(u, v)|^2 = a^2 + b^2$$

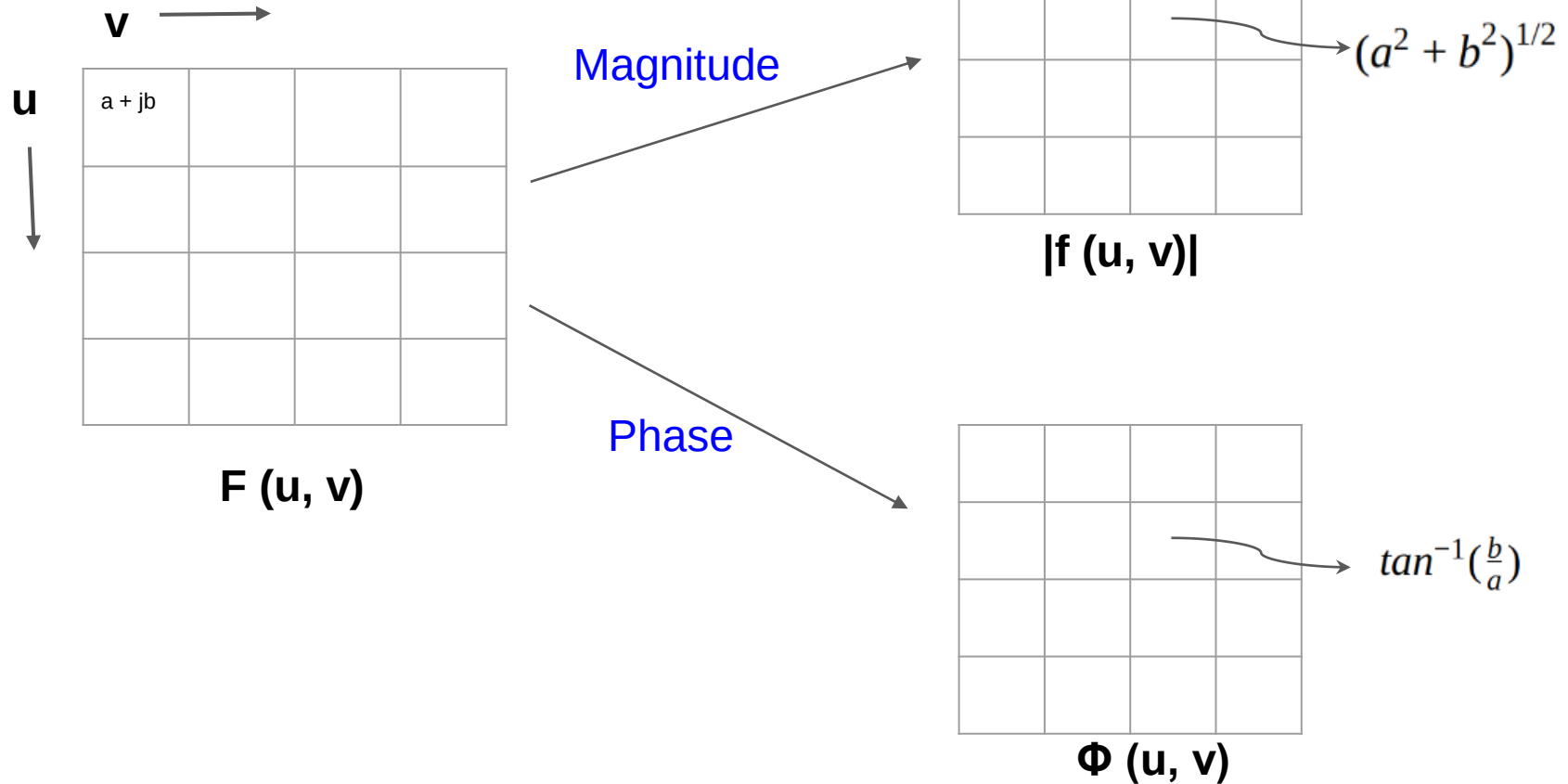


Power Spectrum

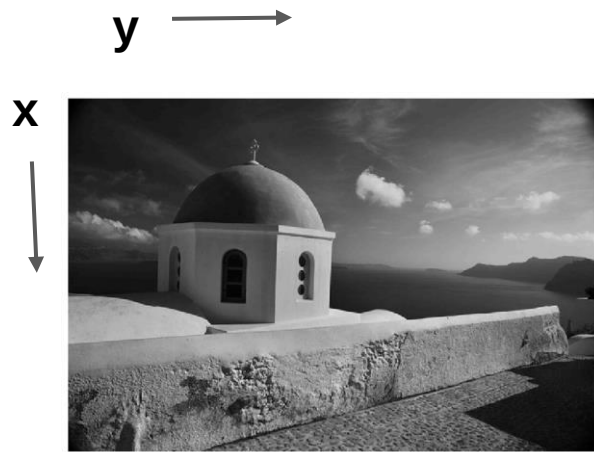
Calculating DFT (Cont.)



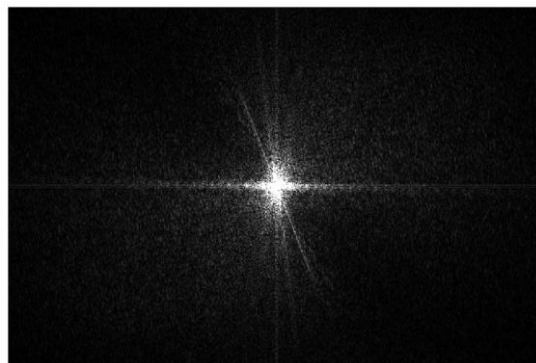
Calculating DFT (Cont.)



Visualizing DFT

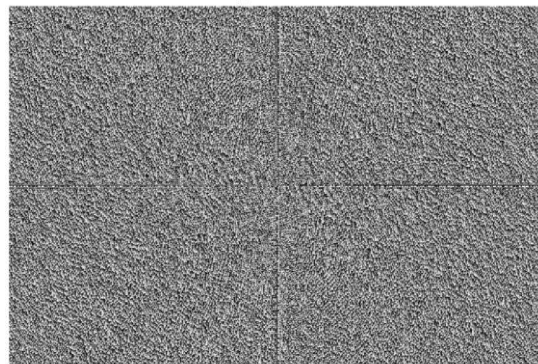


Magnitude



$|f(u, v)|$

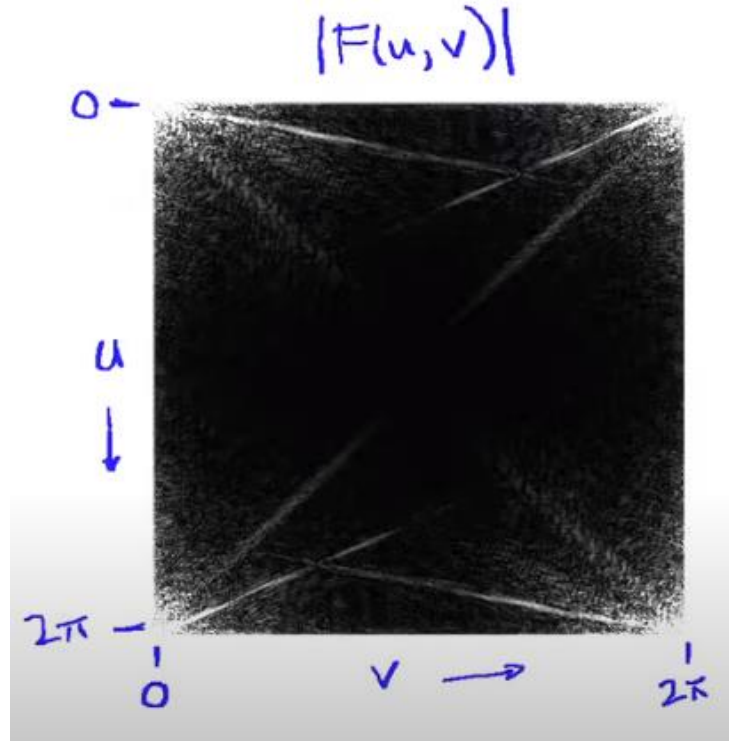
Phase



$\Phi(u, v)$

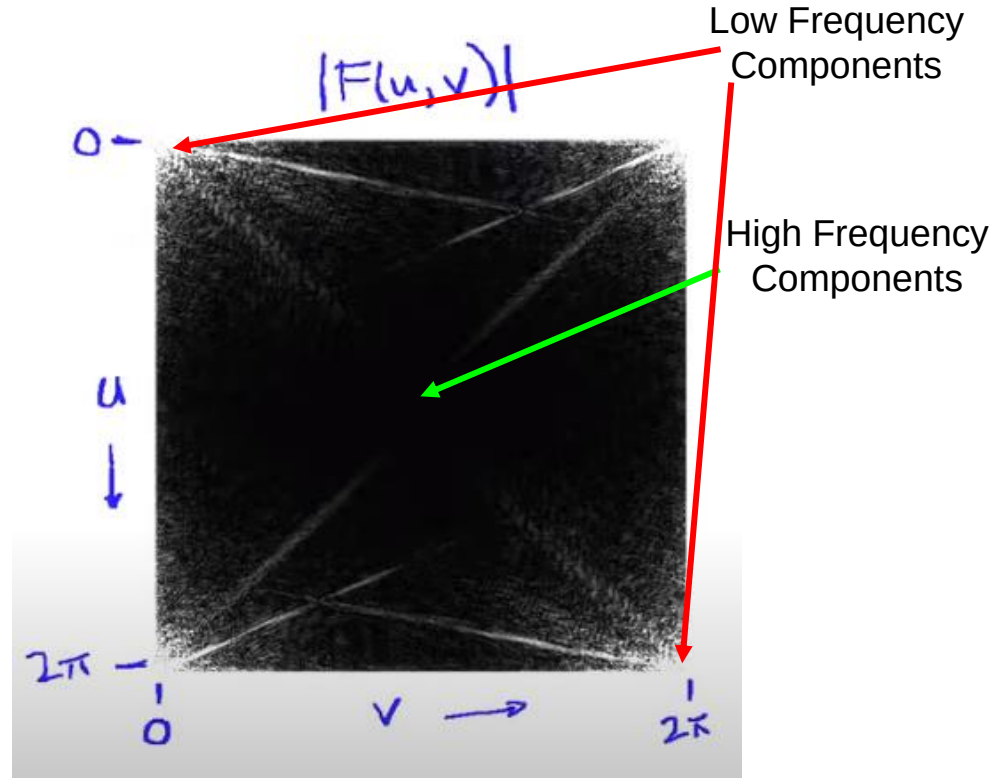
Visualizing DFT (Cont.)

Camerman Image



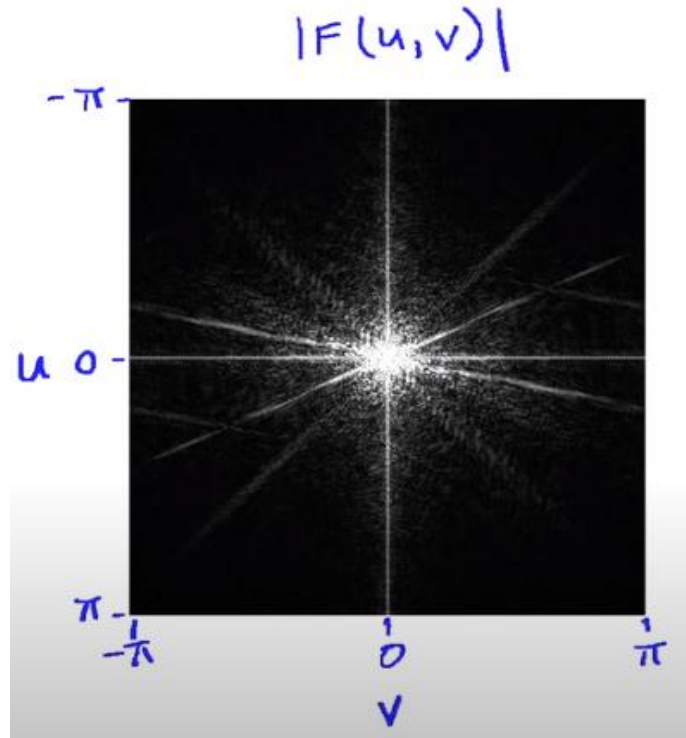
Visualizing DFT (Cont.)

Camerman Image



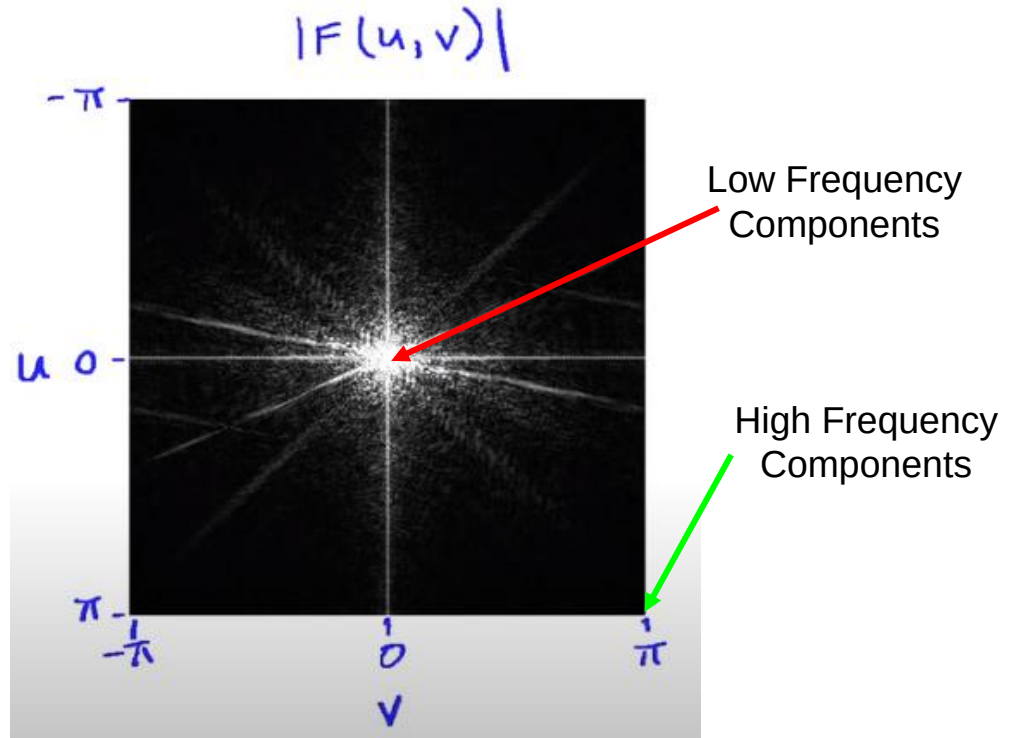
Visualizing DFT (Cont.)

Camerman Image



Visualizing DFT (Cont.)

Camerman Image

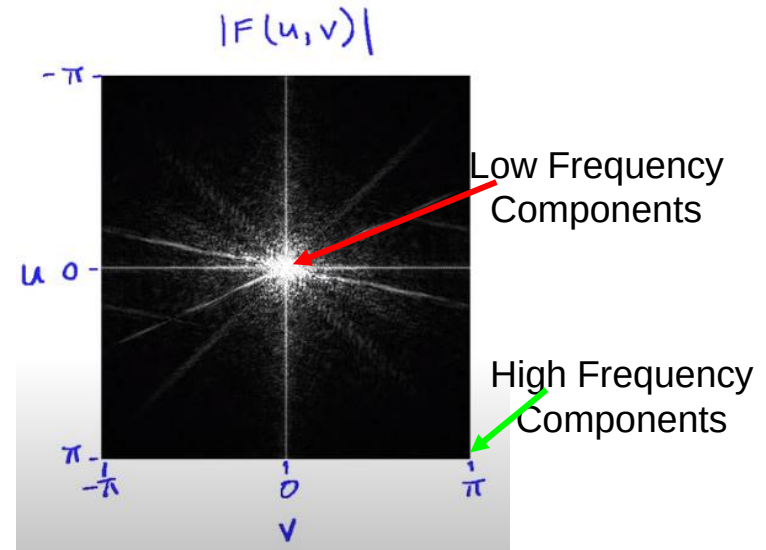


Visualizing DFT (Cont.)

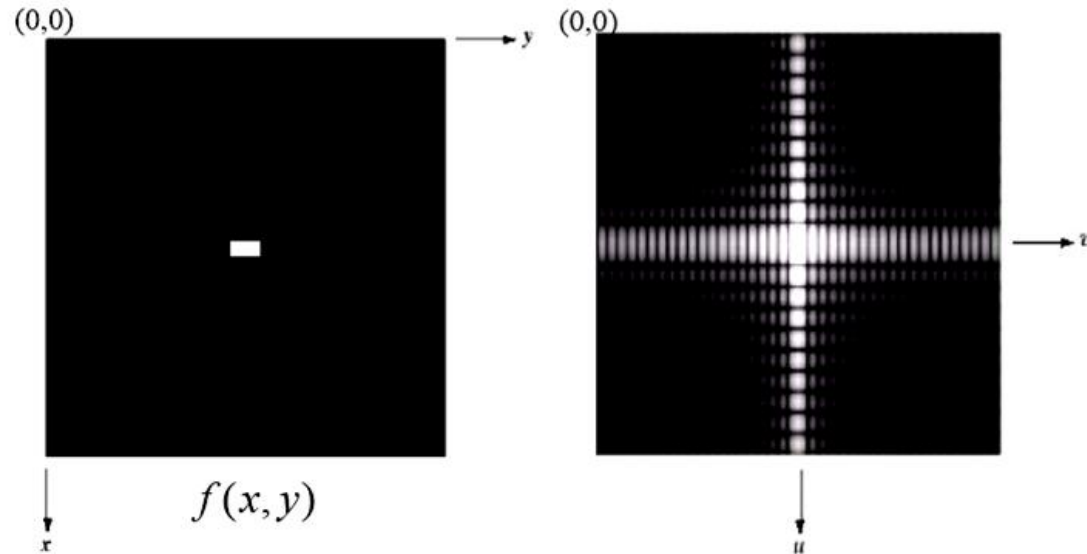
$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

■ $F(0,0) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) :$

- $F(0,0)$ represents the DC-component of the image which corresponds to the average brightness
- Low frequencies
 - close to the center
 - corresponding to slowly varying components of an image
- High frequencies
 - away from the center
 - corresponding to fast gray level changes in an image

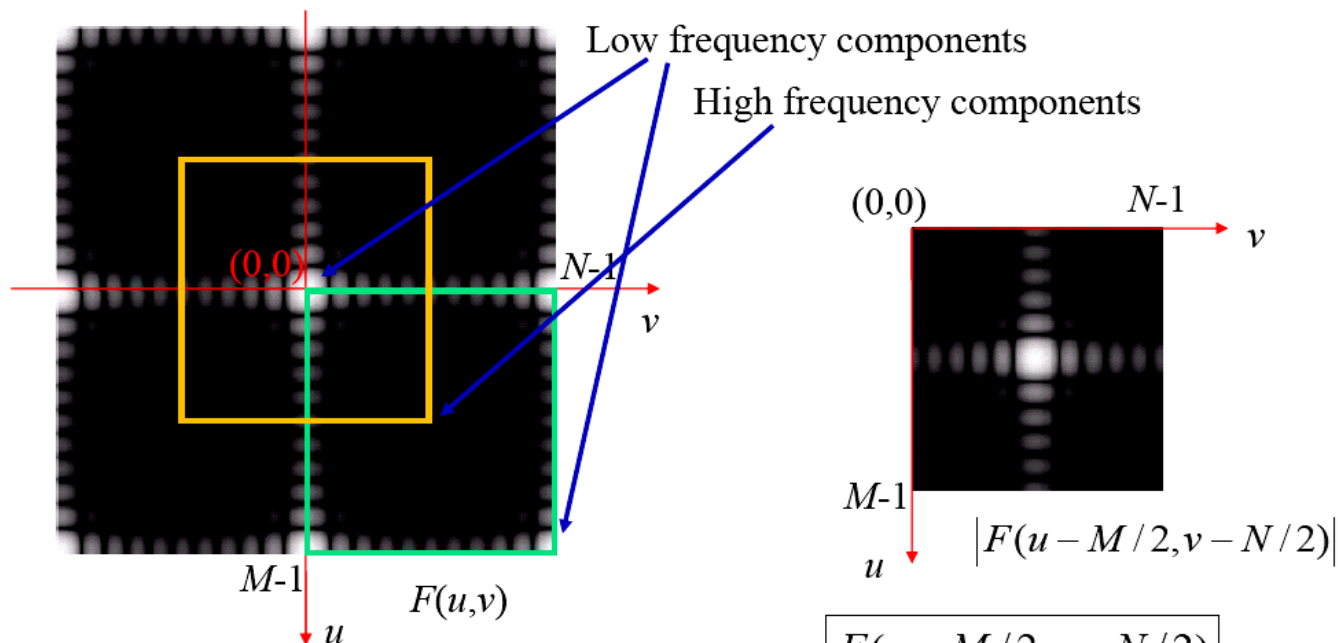


Shifting Origin to the Center



$$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$$

Shifting Origin to the Center (Cont.)



$|F(u,v)| = |F(-u,-v)|$: symmetric;
the spectrum is centered on the origin

$$\boxed{\begin{array}{c} F(u - M/2, v - N/2) \\ \Updownarrow \\ f(x,y)(-1)^{x+y} \end{array}}$$

Translation Property of 2D DFT

For a function $f(x,y)$, the DFT is given as:

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

If instead of $f(x,y)$, we put $f(x,y)e^{j2\pi(u_0x/M + v_0y/N)}$

The DFT will be equal to

$$\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi((u-u_0)x/M + (v-v_0)y/N)} = F(u-u_0, v-v_0)$$

Therefore the Fourier transform pair has the following translation property

$$f(x,y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u-u_0, v-v_0)$$

Similarly $f(x-x_0, y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(ux_0/M + vy_0/N)}$

Translation Property of 2D DFT (Cont.)

If we put $u_0 = M / 2$, and $v_0 = N / 2$,

$$\begin{aligned} e^{j2\pi(u_0x/M+v_0y/N)} &= e^{j\pi(x+y)} \\ &= (-1)^{(x+y)} \end{aligned}$$

This means that

$$f(x,y)(-1)^{(x+y)} \Leftrightarrow F(u-M/2, v-N/2)$$

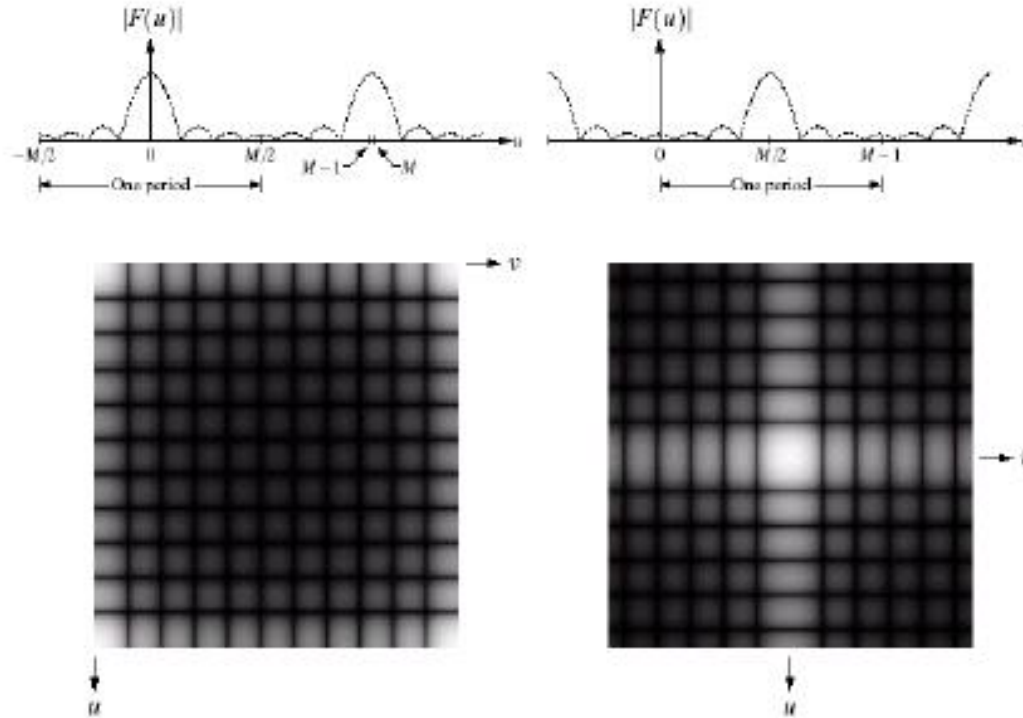
To transform the origin to the center in the transformed image the input is always multiplied by the factor $(-1)^{(x+y)}$

Translation Property of 2D DFT (Cont.)

a b
c d

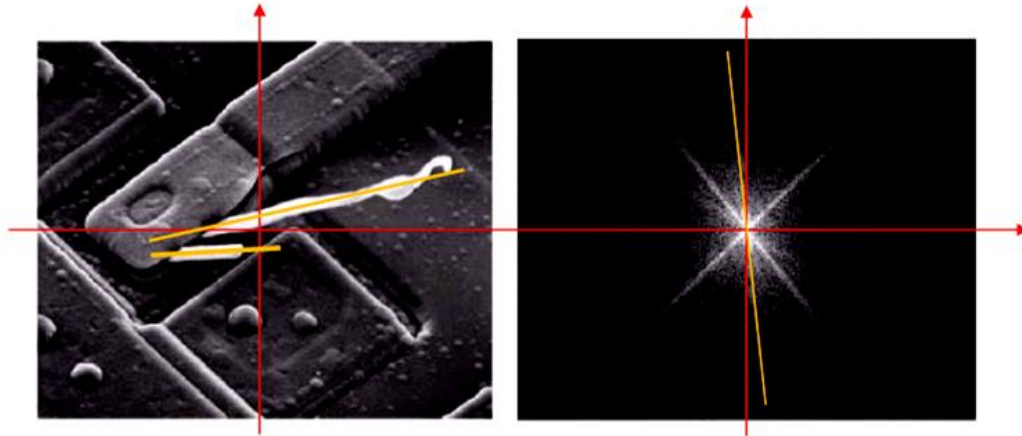
FIGURE 4.34

(a) Fourier spectrum showing back-to-back half periods in the interval $[0, M - 1]$.
(b) Shifted spectrum showing a full period in the same interval.
(c) Fourier spectrum of an image, showing the same back-to-back properties as (a), but in two dimensions.
(d) Centered Fourier spectrum.

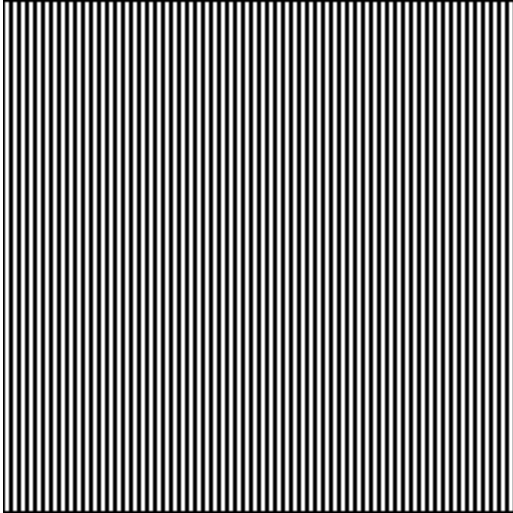


Another Example

- Strong edges run at $\approx \pm 45^\circ$
↑
strong frequency components along $\pm 45^\circ$
- Two white oxide protrusions
↑
vertical component off-axis slightly to the left

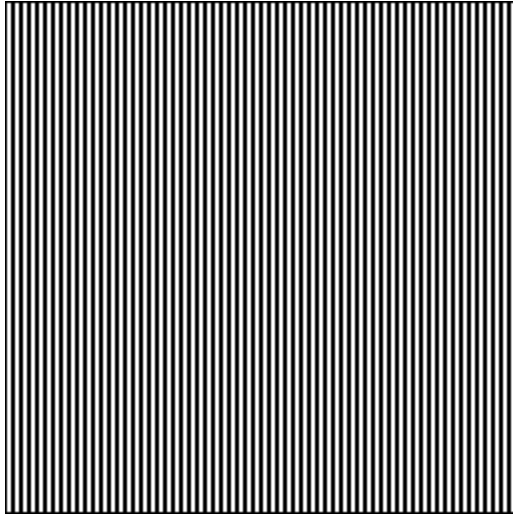


More Examples (Cont.)



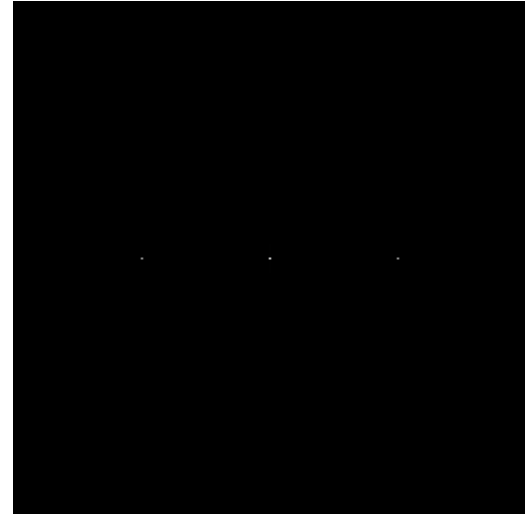
2 Pixel wide vertical stripe

More Examples (Cont.)



DFT

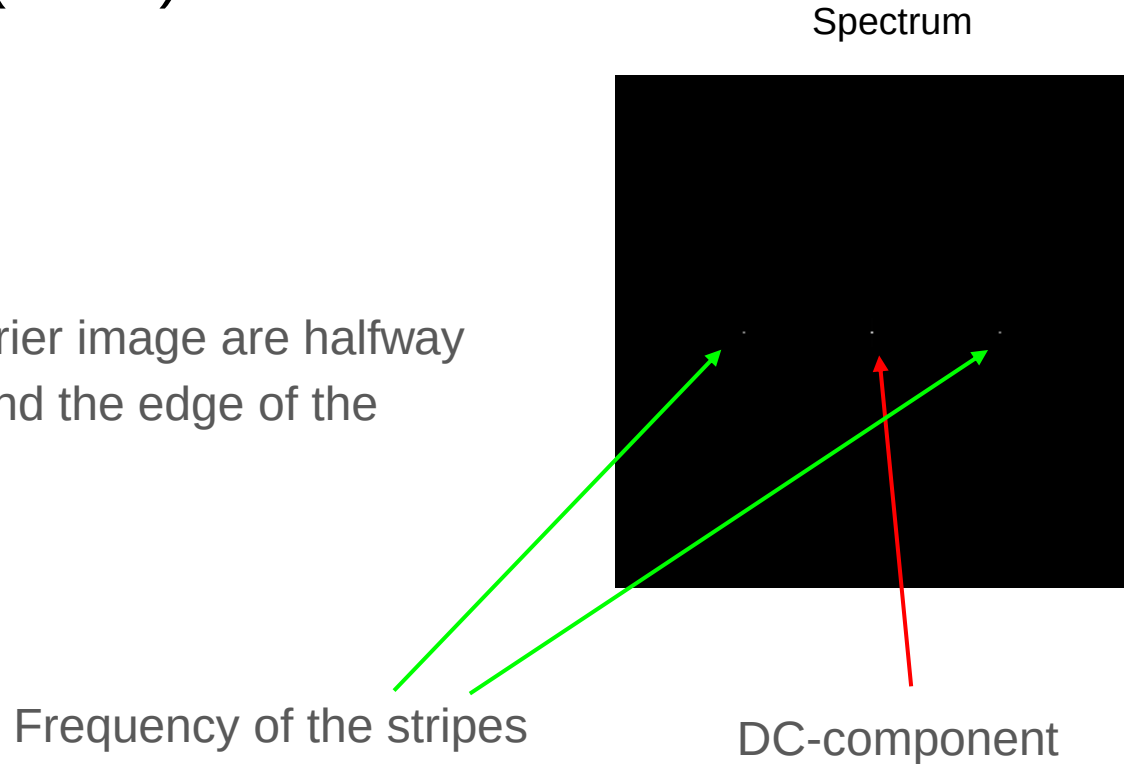
Spectrum



More Examples (Cont.)

$$f = \frac{1}{4 \text{ pixels}} = \frac{f_{\max}}{2}$$

The points in the Fourier image are halfway between the center and the edge of the image

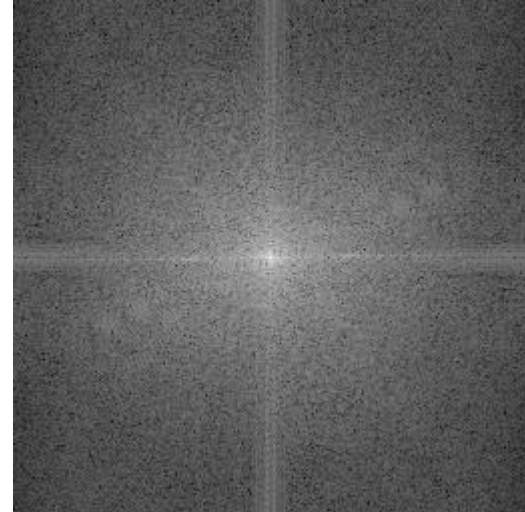


More Examples (Cont.)



DFT

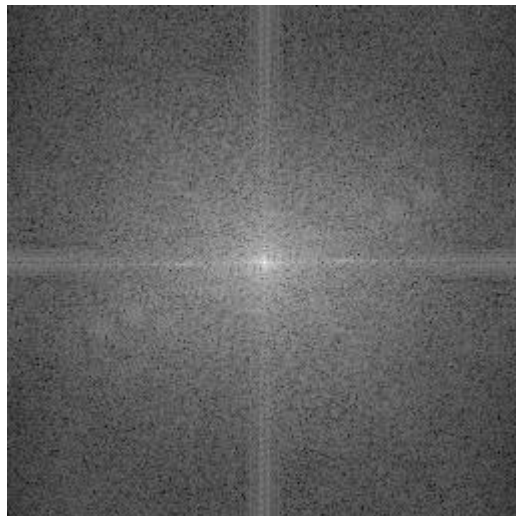
Spectrum



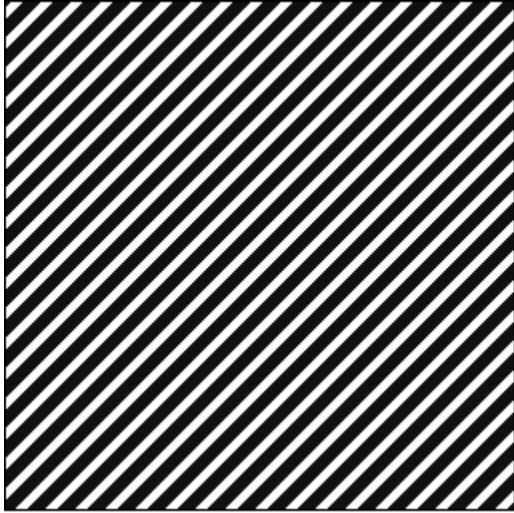
More Examples (Cont.)

- The image contains components of all frequencies
 - their magnitude gets smaller for higher frequencies
 - low frequencies contain more image information than the higher ones
- Two dominating directions in the Fourier image, one passing vertically and one horizontally through the center
 - These originate from the regular patterns in the background of the original image

Spectrum



More Examples (Cont.)

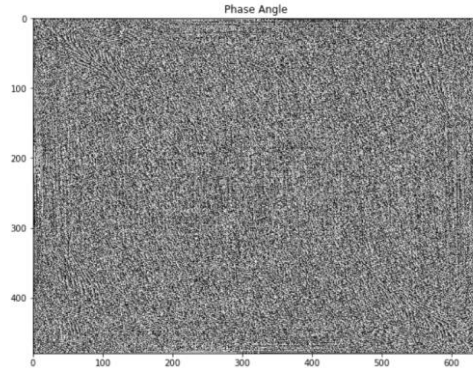
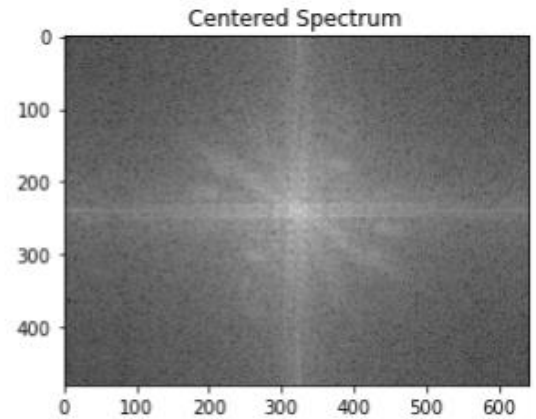
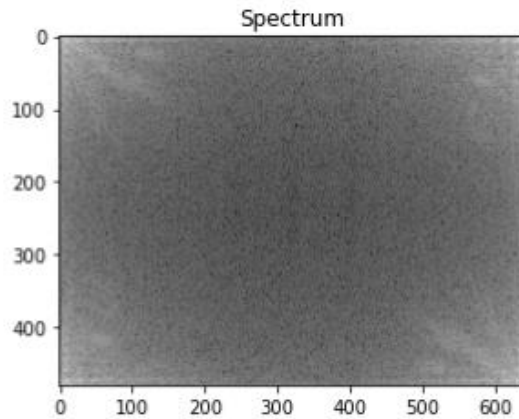
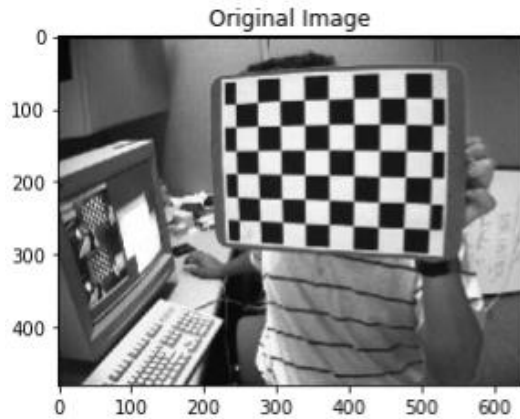


DFT

Spectrum



More Examples (Cont.)

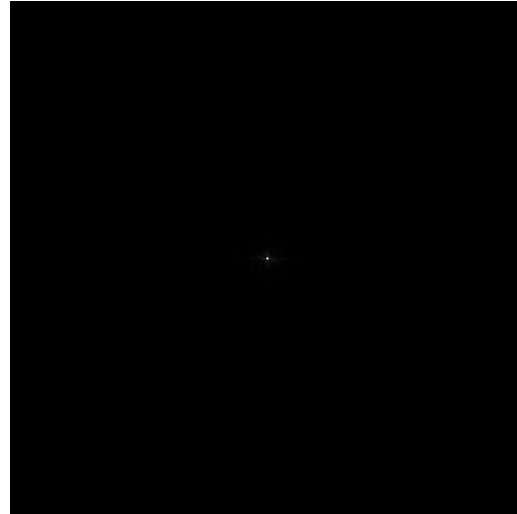


Notes on Showing DFT



DFT

Spectrum



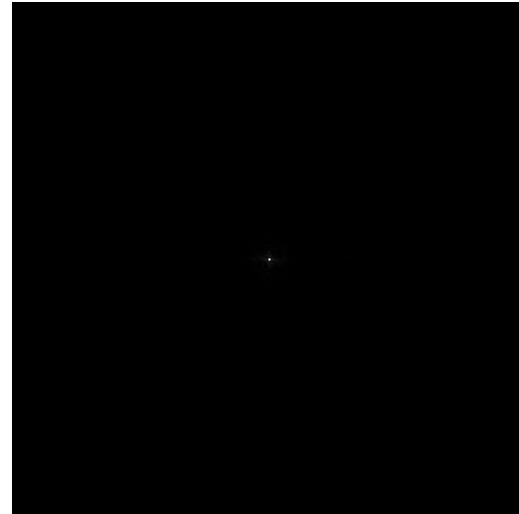
- Only the DC-value is visible
- The dynamic range of the Fourier coefficients (i.e. the intensity values in the Fourier image) is too large to be displayed on the screen
 - all other values appear as black

Notes on Showing DFT



DFT

Spectrum

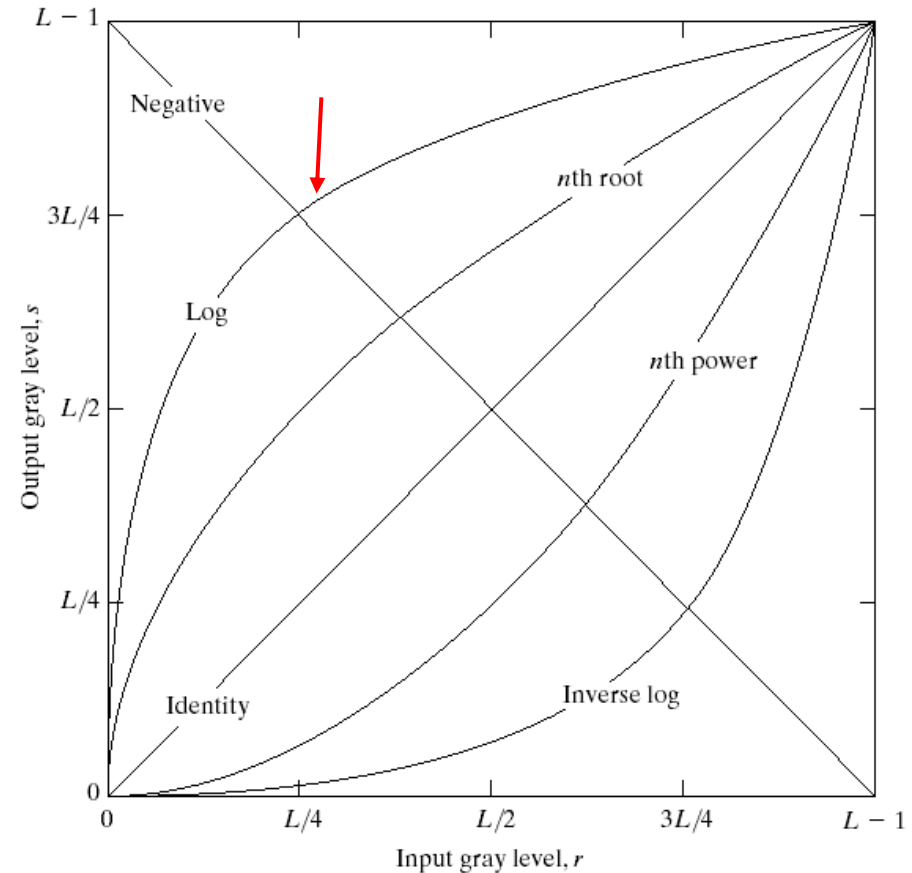


Whis is the solution ??

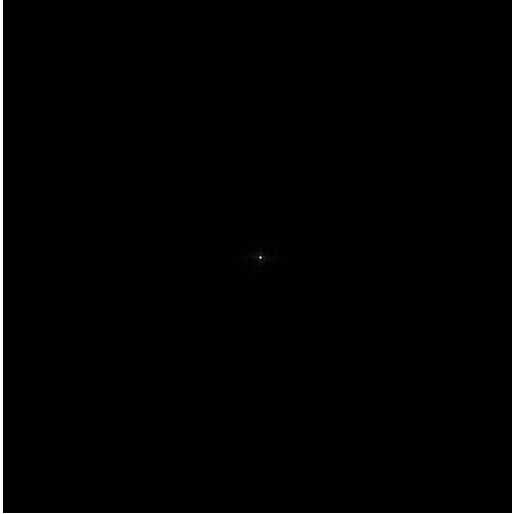
- Only the DC-value is visible
- The dynamic range of the Fourier coefficients (i.e. the intensity values in the Fourier image) is too large to be displayed on the screen
 - all other values appear as black

Notes on Showing DFT

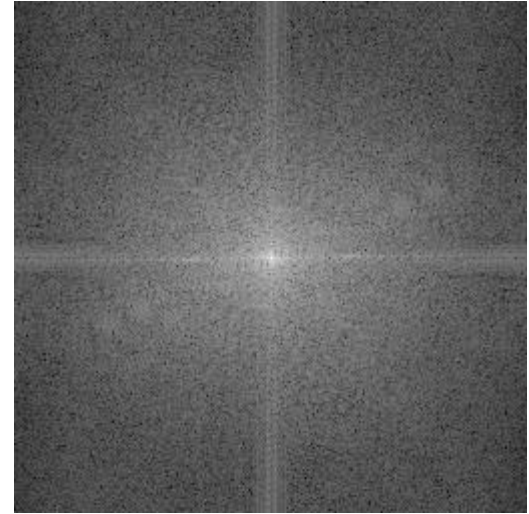
- Log transformation
- $s = c \log(1+r)$
- Expands the dark pixels in the image while compressing the brighter pixels
- Compresses the dynamic range (display of Fourier transform)



Notes on Showing DFT



Log
Transformation

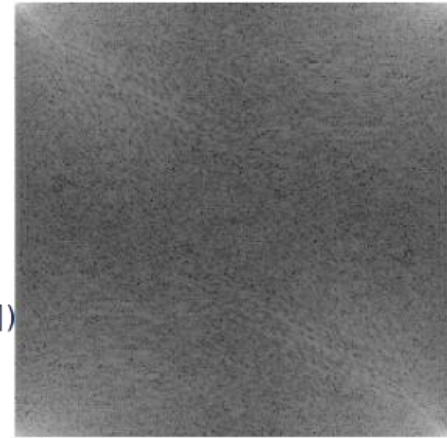


Notes on Showing DFT



Lena 256x256

```
imshow(log(abs(F)), [])
```

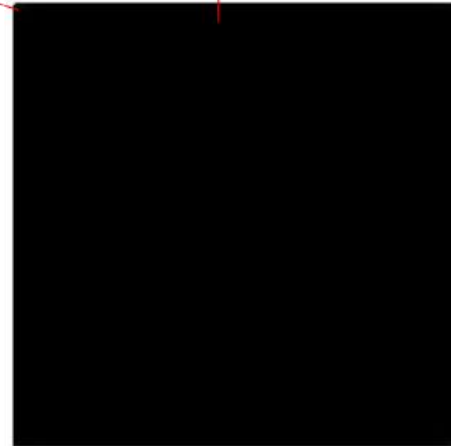


$F(1,1)=7761921$

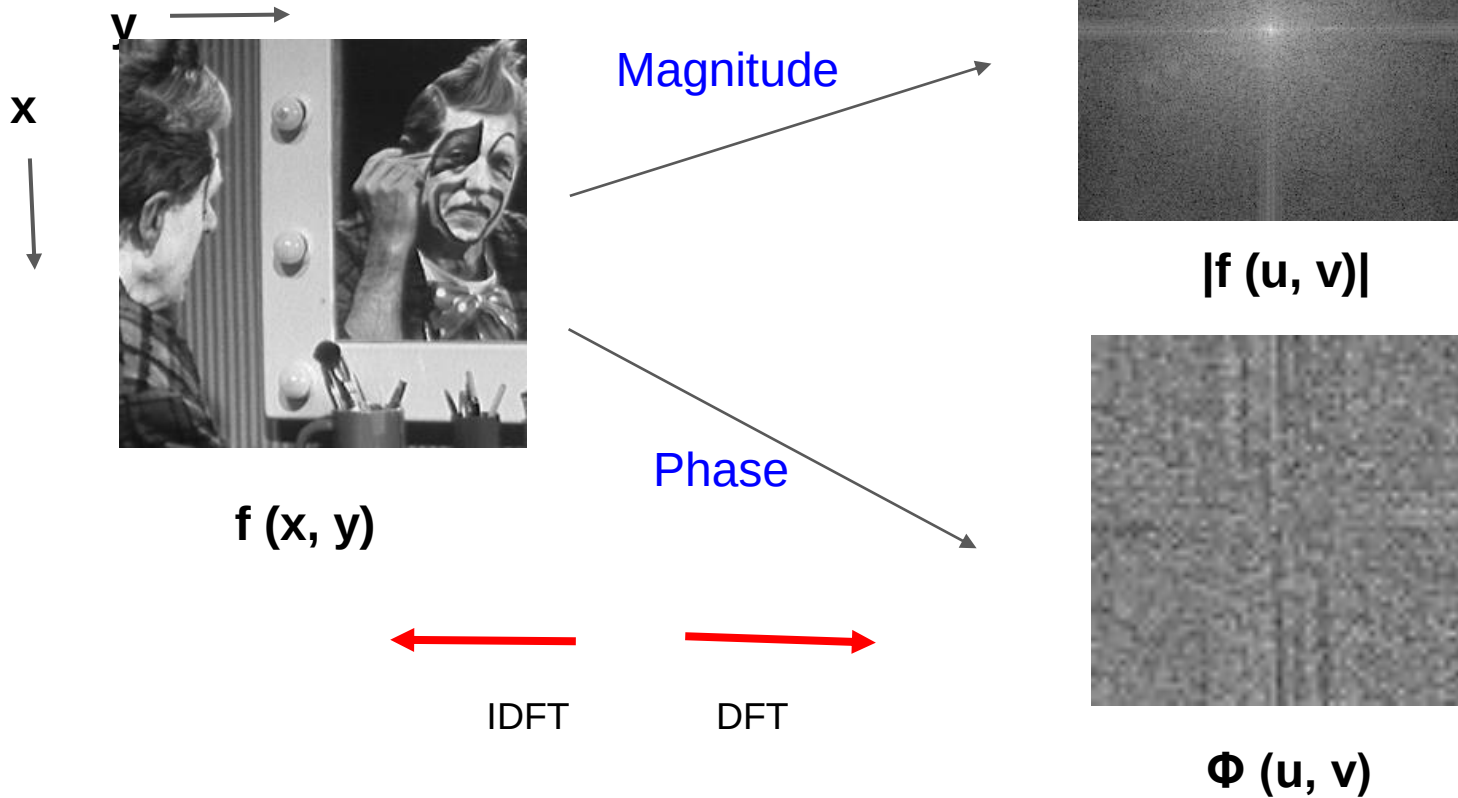
$F(1,127)=334.79+10i$

→

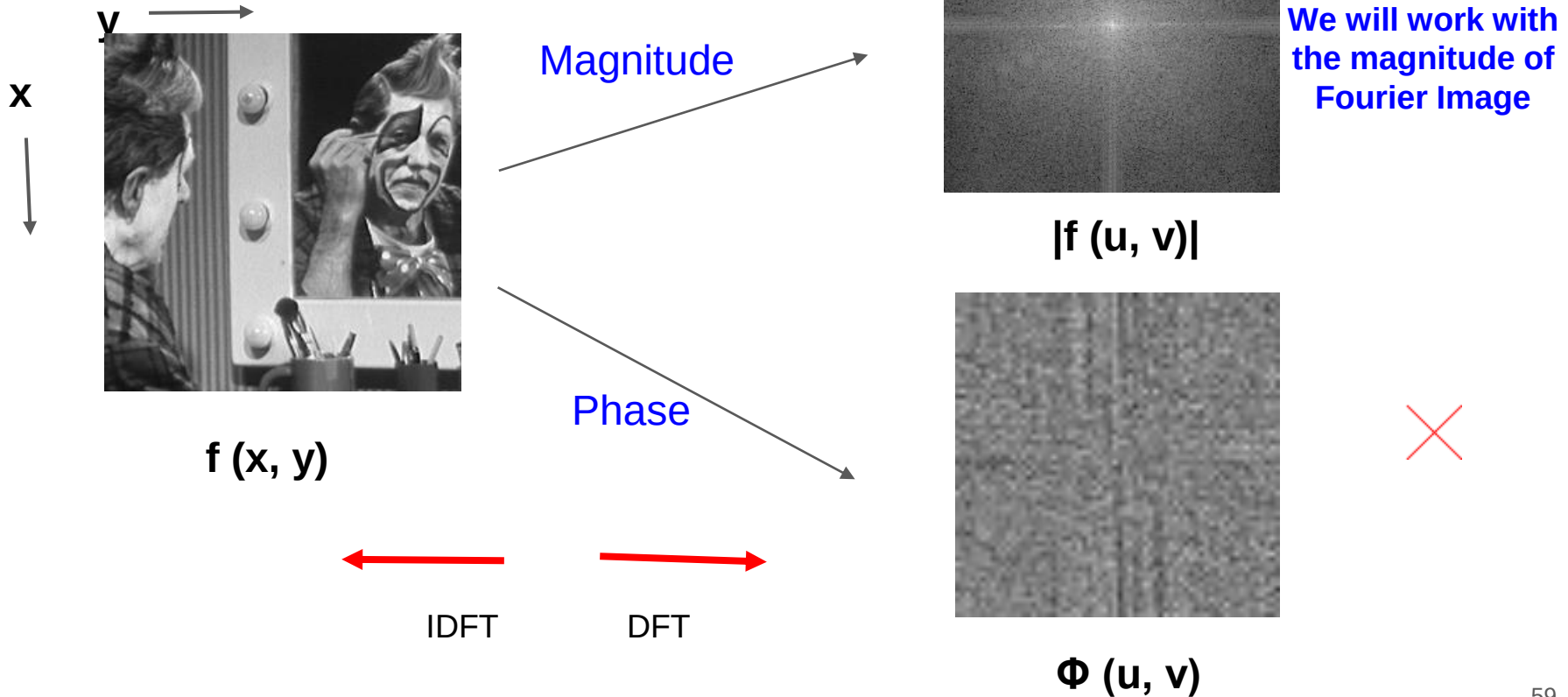
```
F=fft2(I);  
imshow(abs(F), [])
```



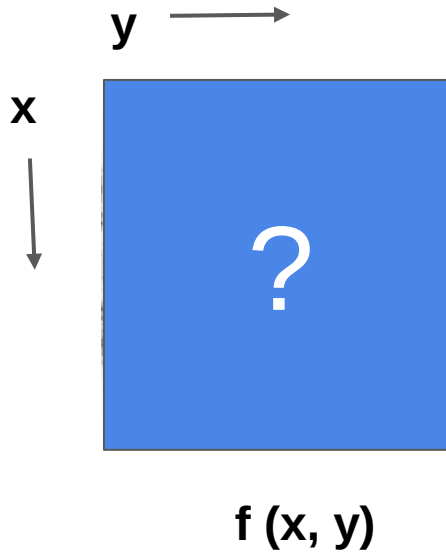
Notes on Showing IDFT



Notes on Showing IDFT



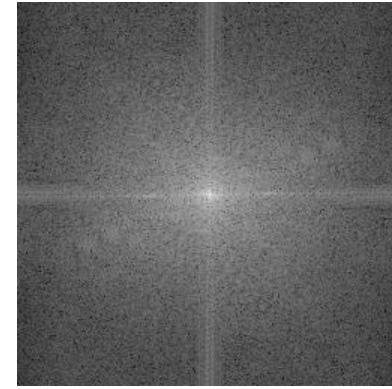
Notes on Showing IDFT



Magnitude

← IDFT

Phase



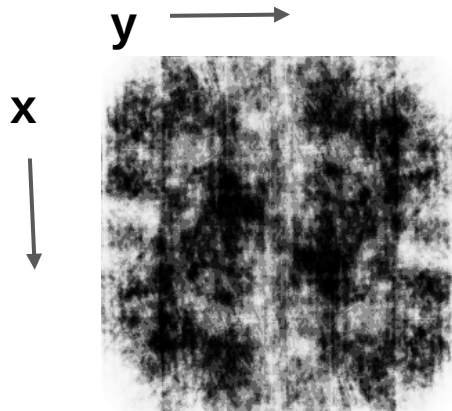
$|f(u, v)|$



$\Phi(u, v)$

What happens if we perform inverse fourier transformation of the magnitude image ignoring phase ??

Notes on Showing IDFT



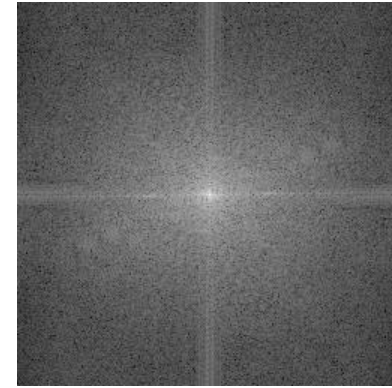
$f(x, y)$

Magnitude



IDFT

Phase



$|f(u, v)|$



$\Phi(u, v)$

- Contains the same frequencies (and amount of frequencies) as the original input image
- Corrupted beyond recognition
- The phase information is crucial to reconstruct the correct image in the spatial domain

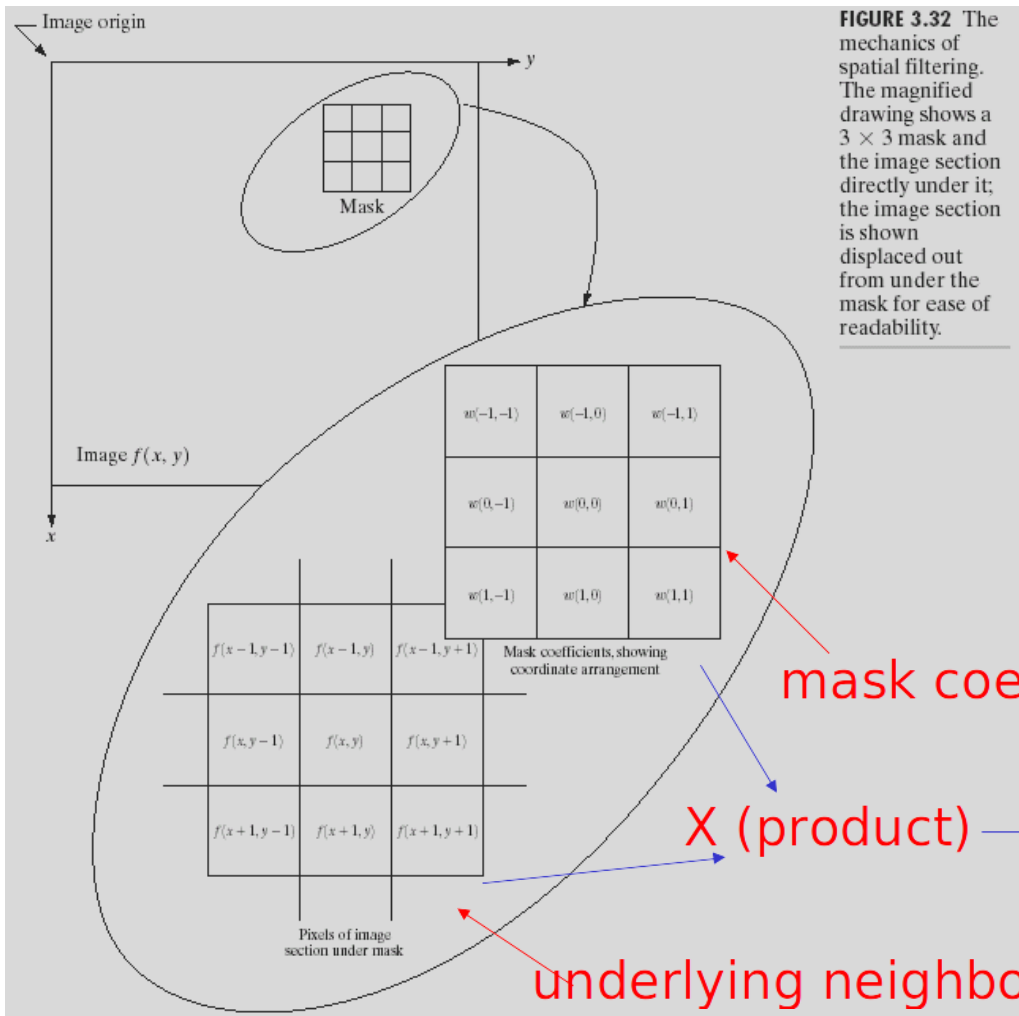


FIGURE 3.32 The mechanics of spatial filtering. The magnified drawing shows a 3×3 mask and the image section directly under it; the image section is shown displaced out from under the mask for ease of readability.

Reviewing Spatial Domain Filtering

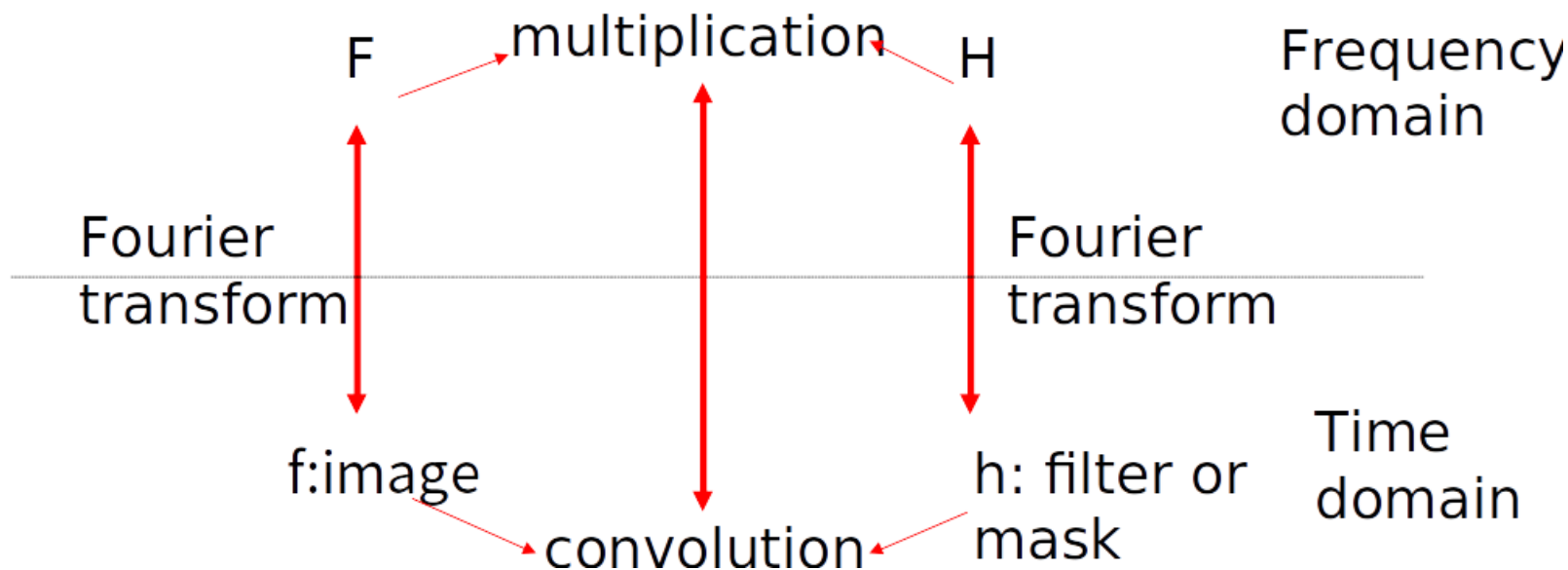
Convolution in 2D Case

$$g(x, y) = f(x, y) \star h(x, y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$$

$$g(x, y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(m - x, n - y)$$

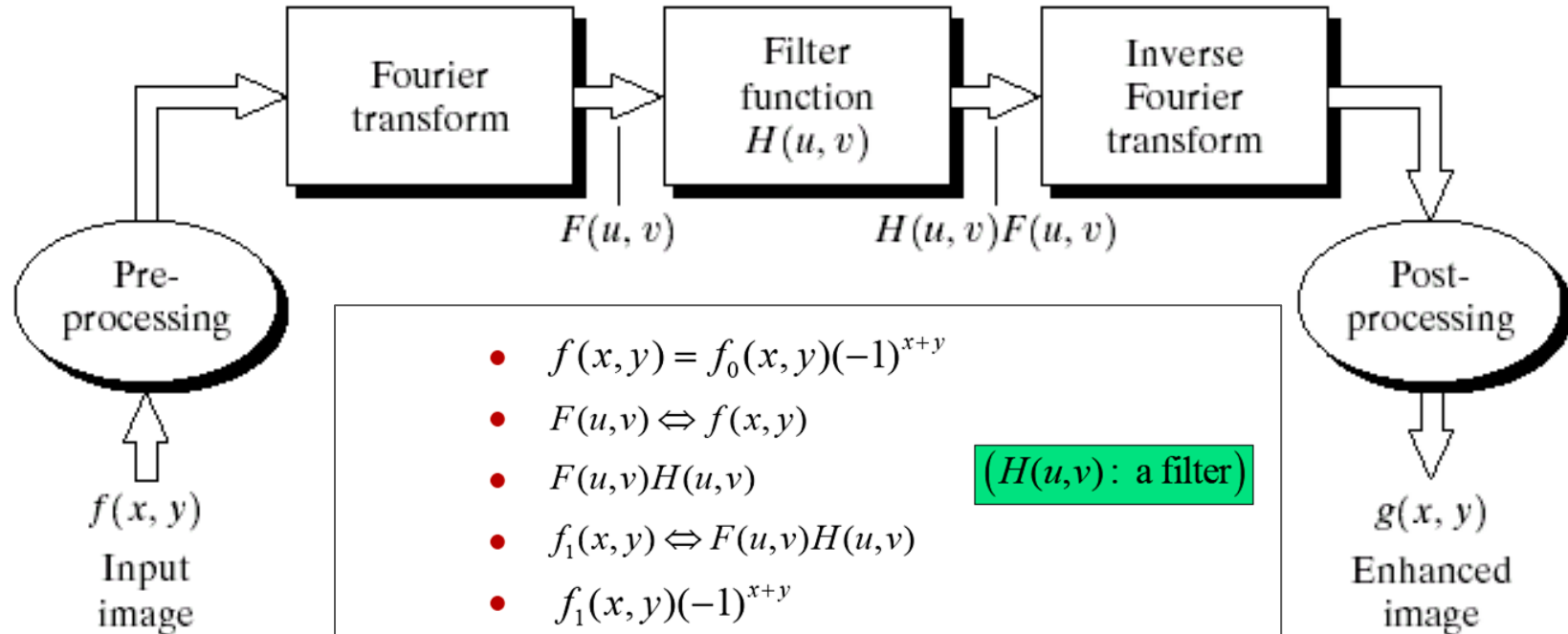
Convolution Theory

$$f(x, y) \star h(x, y) \Leftrightarrow F(u, v) H(u, v)$$



Basic steps for filtering in the frequency domain

Frequency domain filtering operation

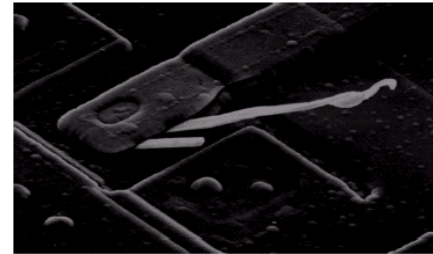
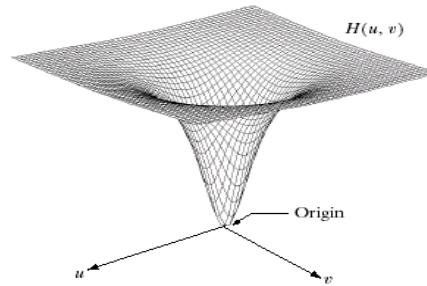
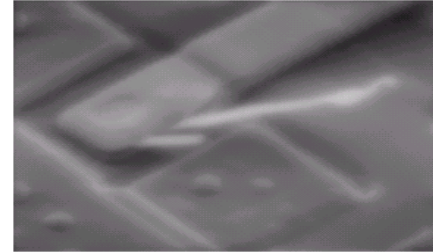
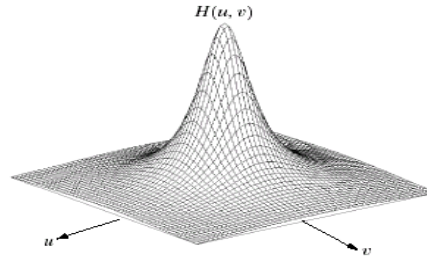
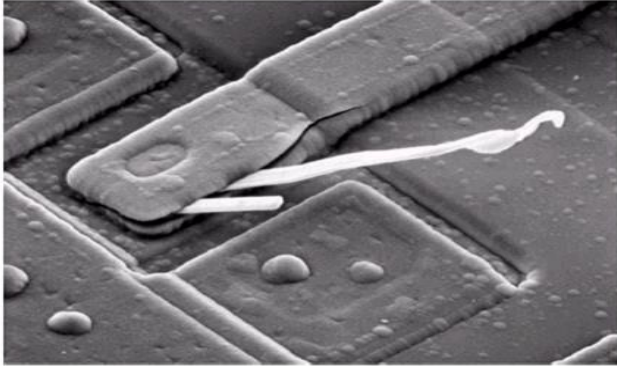


Basic steps for filtering in the frequency domain

1. Multiply the input image by $(-1)^{x+y}$ to center the transform to $u = M/2$ and $v = N/2$
2. Computer $F(u,v)$, the DFT of the image
3. Multiply $F(u,v)$ by a filter function $H(u,v)$
4. Compute the inverse DFT of the result in (3)
5. Obtain the real part of the result in (4)
6. Multiply the result in (5) by $(-1)^{x+y}$ to cancel the multiplication of the input image.

Some Basic Frequency Domain Filters

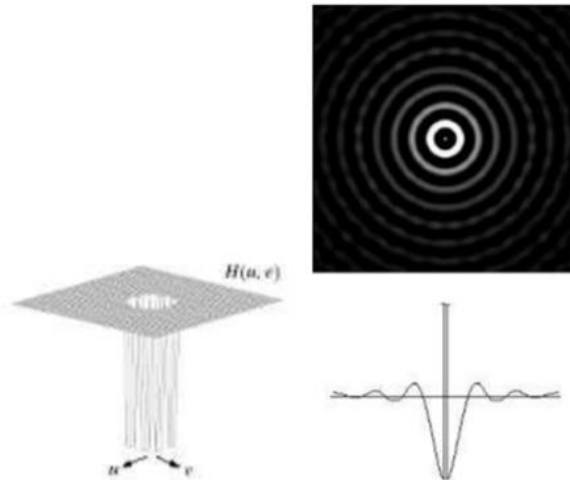
Low Pass Filter



High Pass Filter

Spatial Representation of a Filter

1. Multiply filter $H(u, v)$ by $(-1)^{u+v}$ (centering)
2. Compute the inverse DFT
3. Multiply the real part of the inverse DFT by $(-1)^{x+y}$



Smoothing frequency-domain filters

- Smoothing is achieved in the frequency domain by dropping out the high frequency components (details, noise, ...)
- The basic model for filtering is:
 - $G(u,v) = H(u,v)F(u,v)$
- where $F(u,v)$ is the Fourier transform of the image being filtered and $H(u,v)$ is the filter transform function
- *Low pass filters* – only pass the low frequencies, drop the high ones

Smoothing frequency-domain filters (Cont.)

- ❑ Ideal low-pass filter

- ❑ Butterworth filter

- ❑ Gaussian filter



More smooth in the edge of
cut-off frequency

Ideal Low-pass Filter

- Sharp cut-off frequency

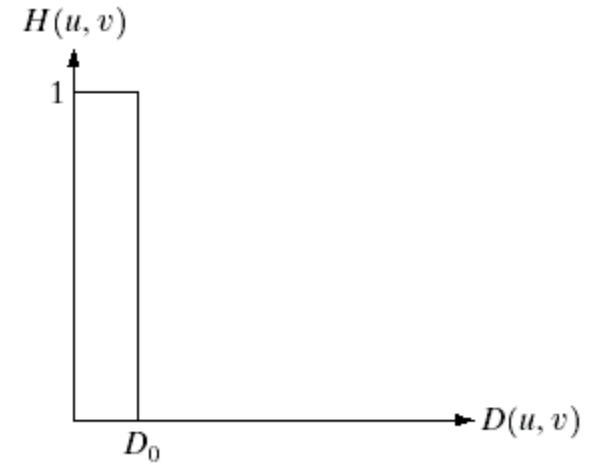
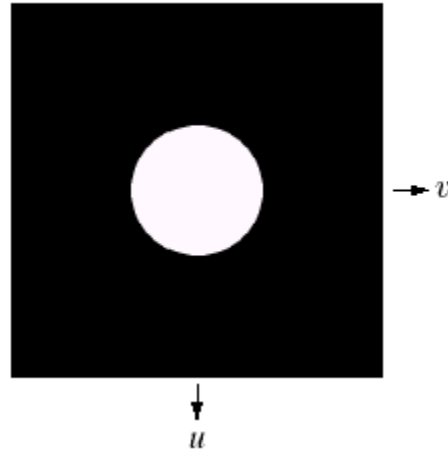
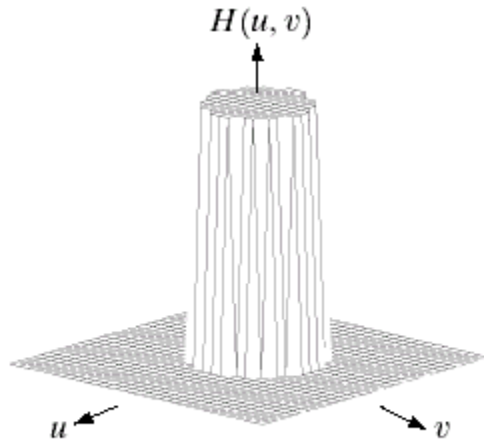
$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) \geq D_0 \end{cases}$$

- D_0 : nonnegative quantity
- $D(u, v)$: distance from a point (u, v) to the origin
- origin: $(\frac{M}{2}, \frac{N}{2})$ (centered)

Then,

$$D(u, v) = \sqrt{(u - \frac{M}{2})^2 + (v - \frac{N}{2})^2}$$

Ideal Low-pass Filter (Cont.)



Cut-off frequency

Ideal Low-pass Filter (Cont.)

- ILPF can not be realized in electronic components, but can be implemented in a computer
- Decision of cut-off freq.?
 - Compute circles that enclose specific amount of total image power
 - Measure the percentage of image power within the low freq.

$$\alpha = 100 \left[\sum_u \sum_v P(u, v) / P_T \right]$$

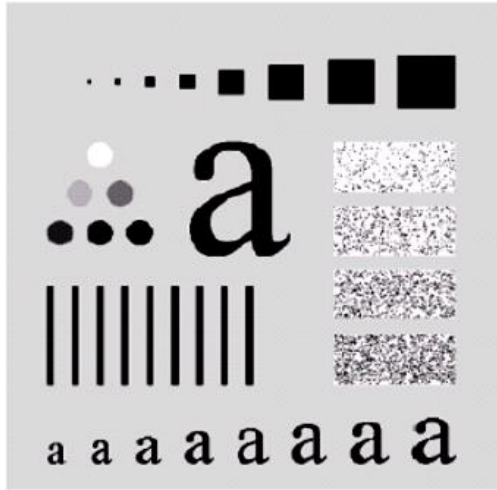
- Total image Power

$$P_T = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} P(u, v)$$

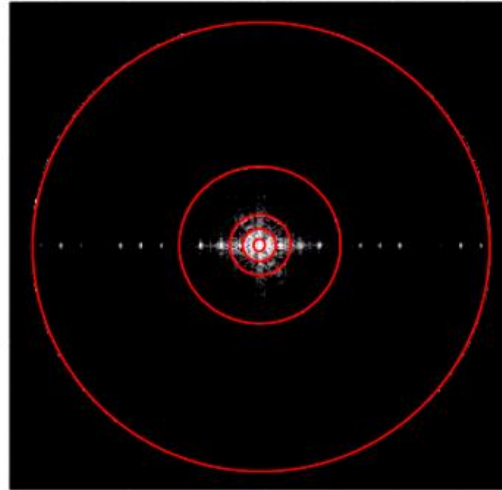
$$P(u, v) = |F(u, v)|^2 = R^2(u, v) + I^2(u, v)$$

ILPF: Distribution of Image Power

$\alpha = 92.0, 94.6, 96.4, 98, 99.5$
% total power



500 × 500



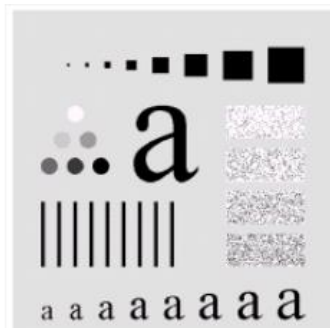
$r = 5, 15,$
 $30, 80, 230$

Spectrum of the left with
5 lowpass filters' radii superimposed

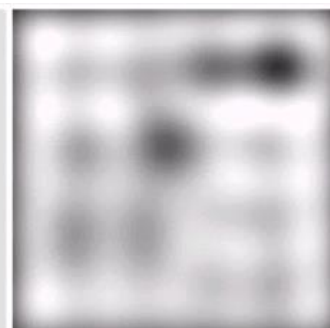
Result of ILPF

Blurring &
Ringing

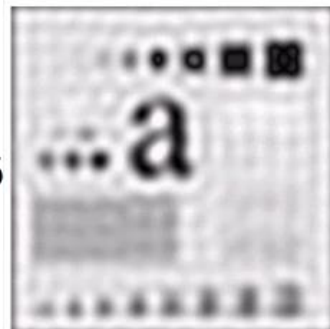
original



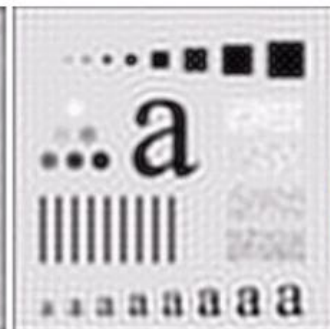
$\alpha=92$
 $D_0=5$



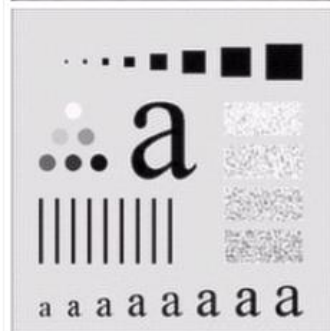
$\alpha=94.6$
 $D_0=15$



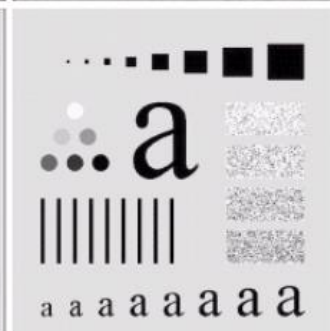
$\alpha=96.4$
 $D_0=30$



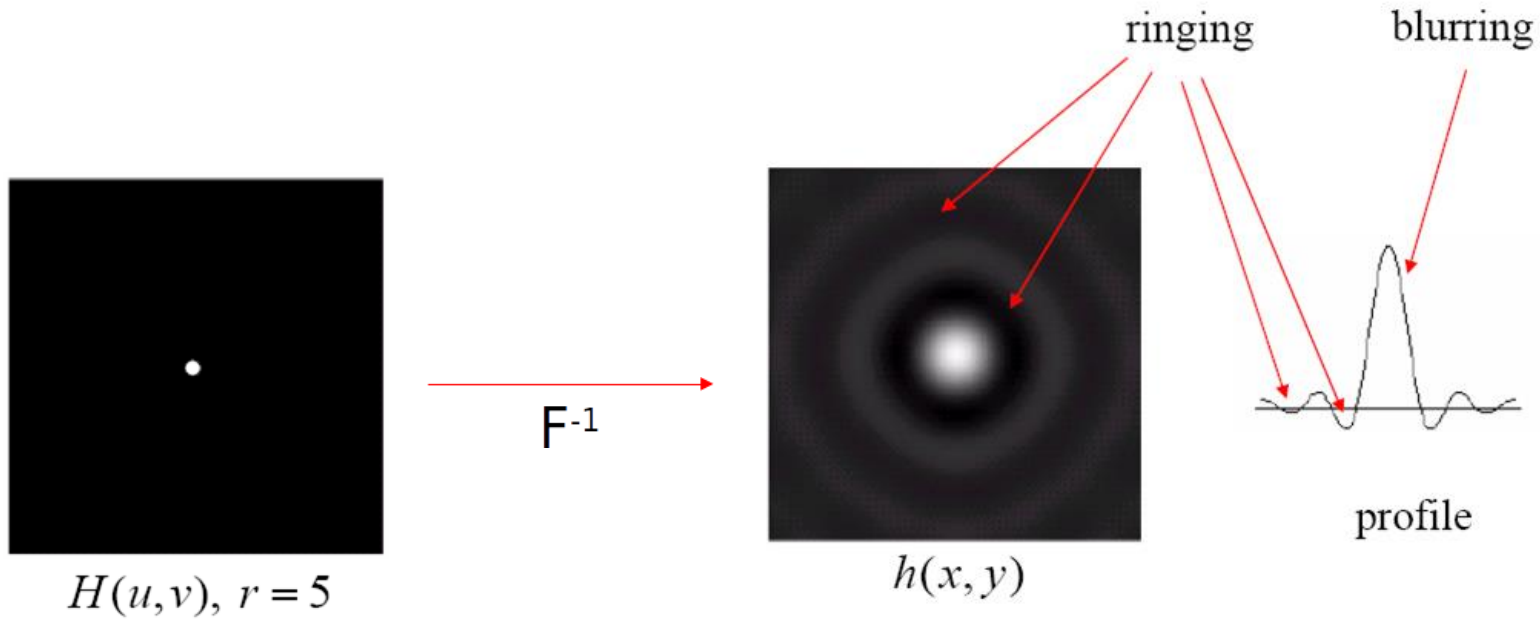
$\alpha=98$
 $D_0=80$



$\alpha=99.5$
 $D_0=230$



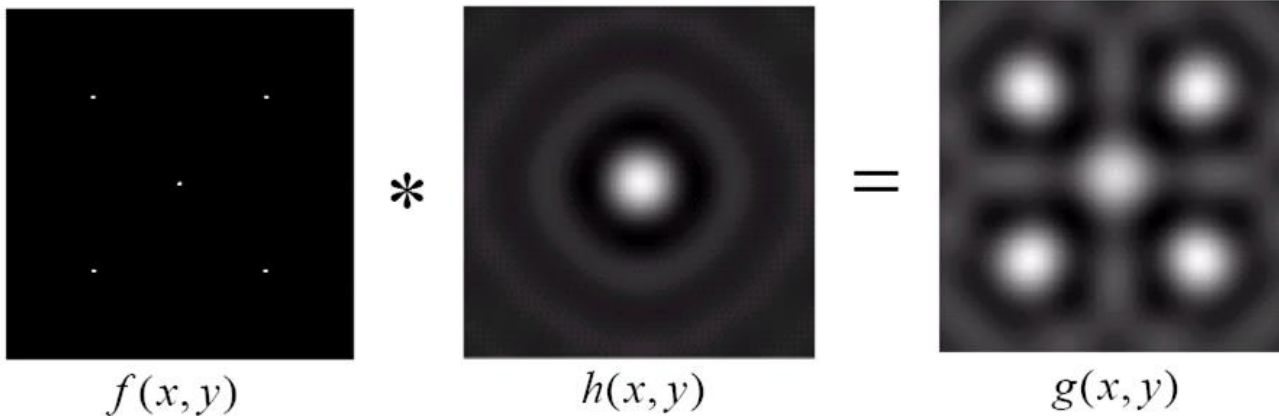
Why Blurring and Ringing?



ILPF: Ringing

$$G(u,v) = F(u,v)H(u,v), \quad g(x,y) = f(x,y) * h(x,y)$$

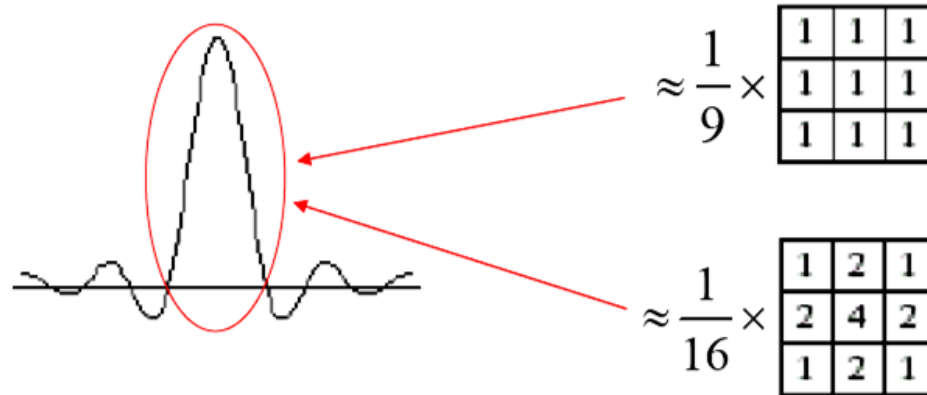
$$g(x,y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)h(x-m, y-n)$$



ILPF: Blurring

$$G(u,v) = F(u,v)H(u,v), \quad g(x,y) = f(x,y) * h(x,y)$$

$$g(x,y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)h(x-m,y-n)$$



Butterworth low-pass filters

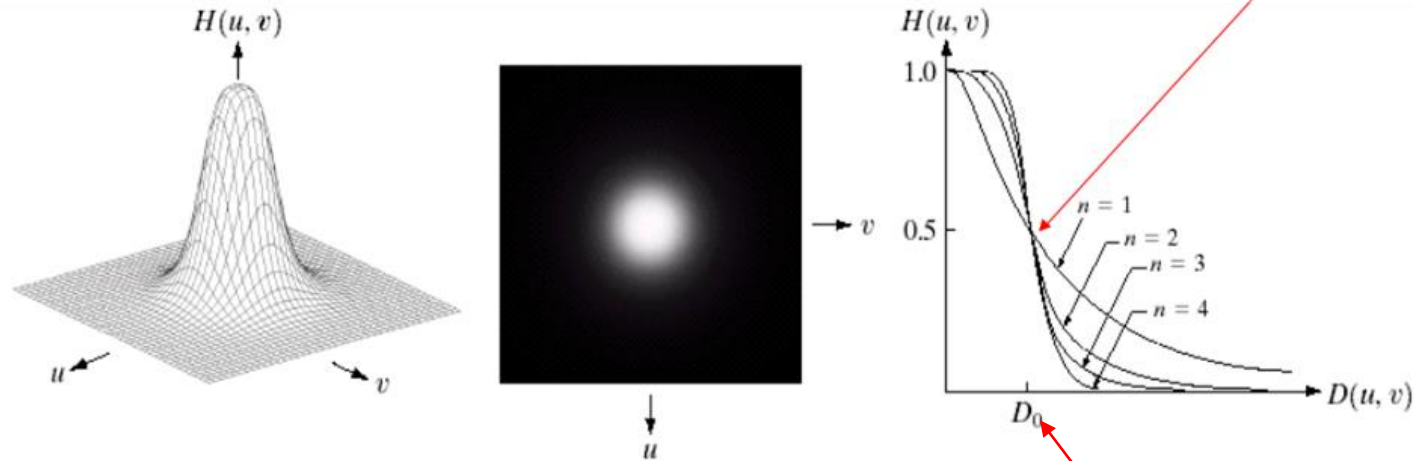
- BLPF of order n , with a cutoff frequency distance D_0 is defined as

$$H(u, v) = \frac{1}{1 + [\frac{D(u, v)}{D_0}]^{2n}}$$

$$D(u, v) = \sqrt{(u - \frac{M}{2})^2 + (v - \frac{N}{2})^2}$$

- no clear cutoff between passed and filtered frequencies

Butterworth low-pass filters



$H=0.5$ when
 $D(u,v)=D_0$

Cut-off frequency

Butterworth low-pass filters

$n = 2$

Blurring &
No visible ringing



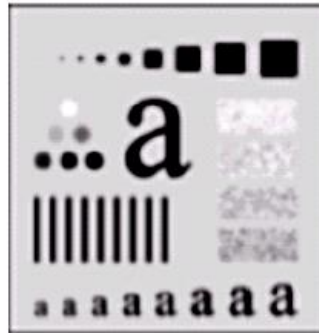
Original image



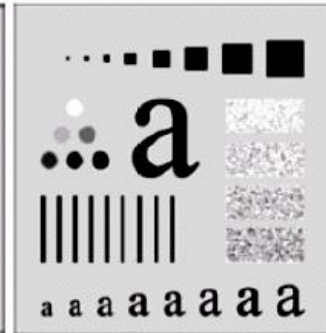
Result with $D_0=5$



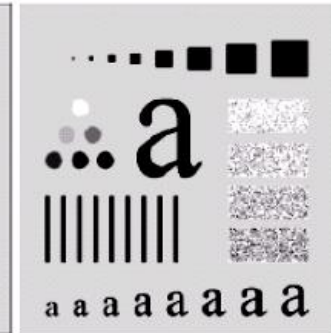
Result with $D_0=15$



Result with $D_0=30$

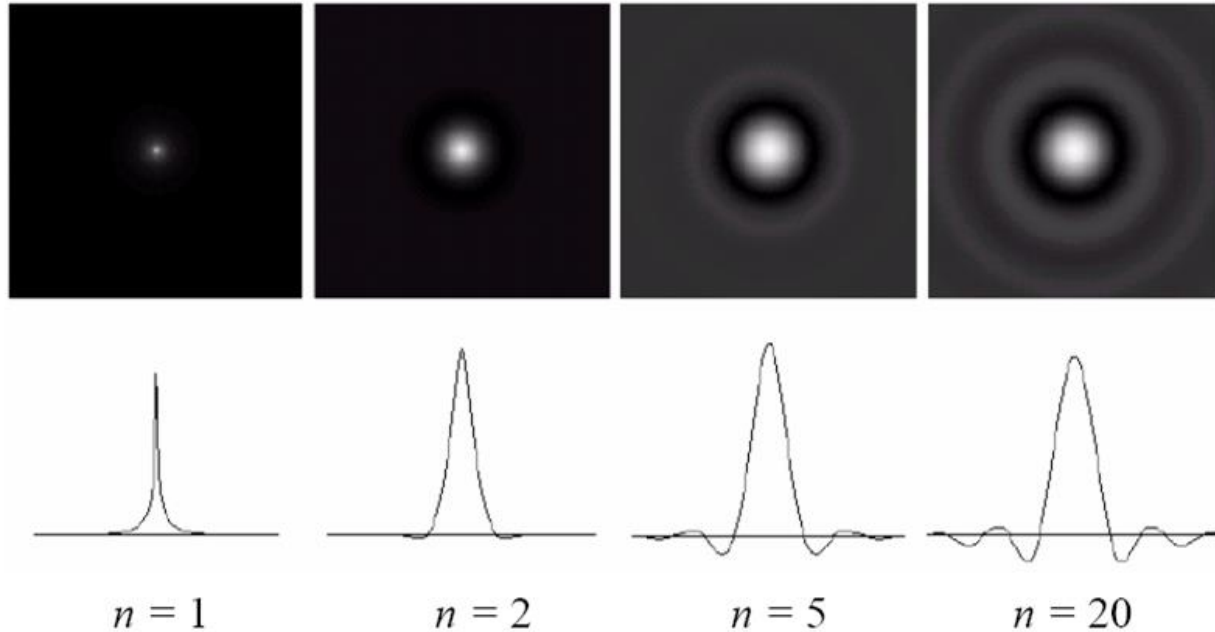


Result with $D_0=80$



Result with $D_0=230$

Spatial Butterworth Low-pass Filters



no ringing effect

little ringing effect

ringing effect

Gaussian Low-pass Filters

- The form of a gaussian lowpass filter GLPF in 2D is:

$$H(u, v) = e^{\frac{-D^2(u, v)}{2\sigma^2}}$$

$$D(u, v) = \sqrt{(u - \frac{M}{2})^2 + (v - \frac{N}{2})^2}$$

- The inverse Fourier transform of a GLPF is also a Gaussian
- A spatial Gaussian filter will have no ringing

Gaussian Low-pass Filters

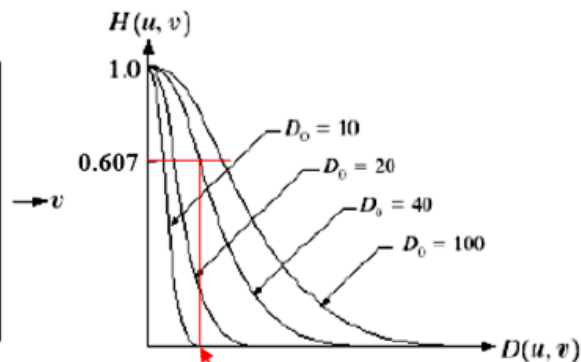
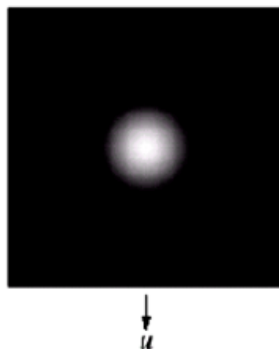
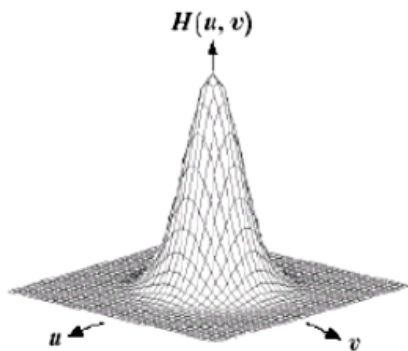
- σ : measure of the spread of the Gaussian curve

- Let $\sigma = D_0$, then:

$$H(u, v) = e^{\frac{-D^2(u, v)}{2D_0^2}}$$

$$H(D(u, v) = 0) = 1$$

$$H(D(u, v) = D_0) = e^{-0.5} = 0.607$$



D_0 : cutoff frequency

$$D = D_0, H = 0.607$$

Gaussian Low-pass Filters

Blurring &
No ringing



Original image



Result with $D_0=5$



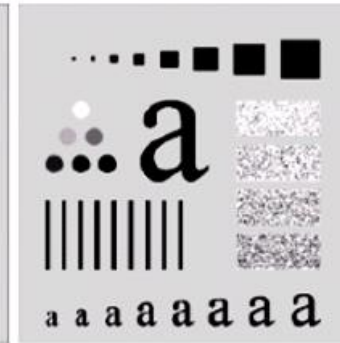
Result with $D_0=15$



Result with $D_0=30$

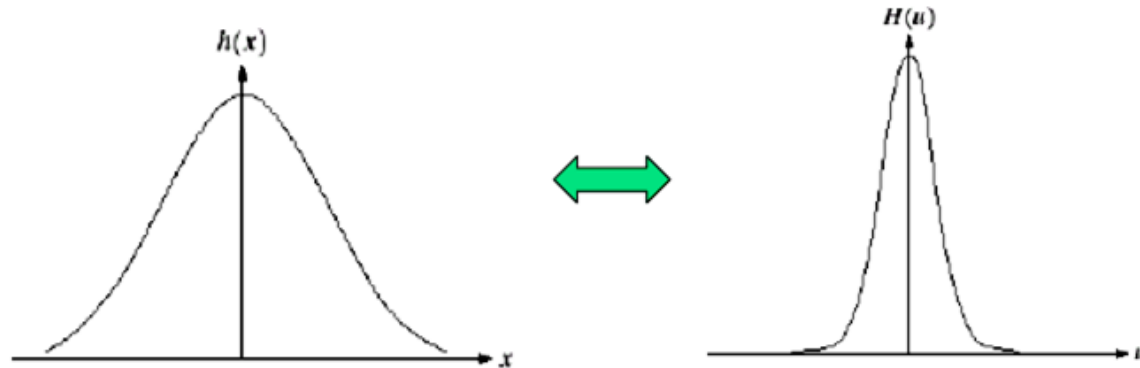


Result with $D_0=80$



Result with $D_0=230$

Gaussian Low-pass Filters



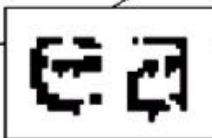
$$\sqrt{2\pi}D_0Ae^{-2\pi^2D_0^2x^2} \Leftrightarrow Ae^{-u^2/2D_0^2}$$

$h(x) > 0$: never cause ringing effect

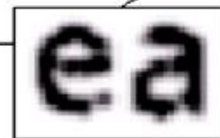
Practical Application 01

(a) Sample text of poor resolution (note broken characters in magnified view).
(b) Result of filtering with a GLPF (broken character segments were joined).

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

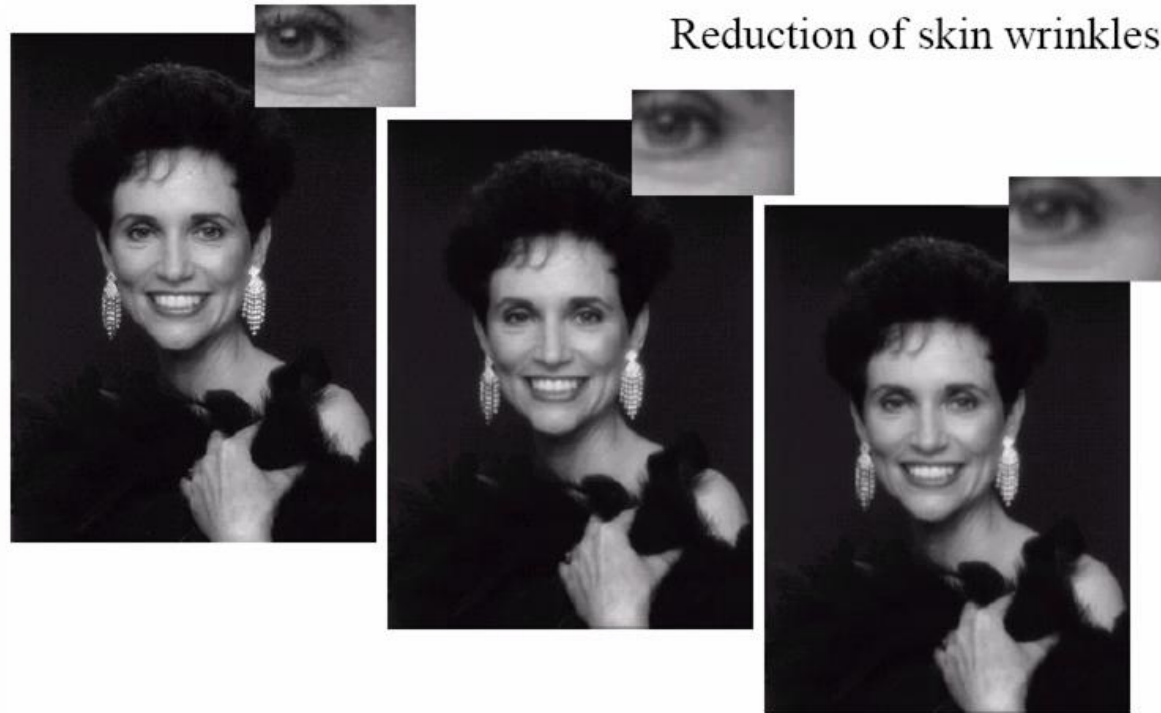


Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



$$D_0 = 80$$

Practical Application 02

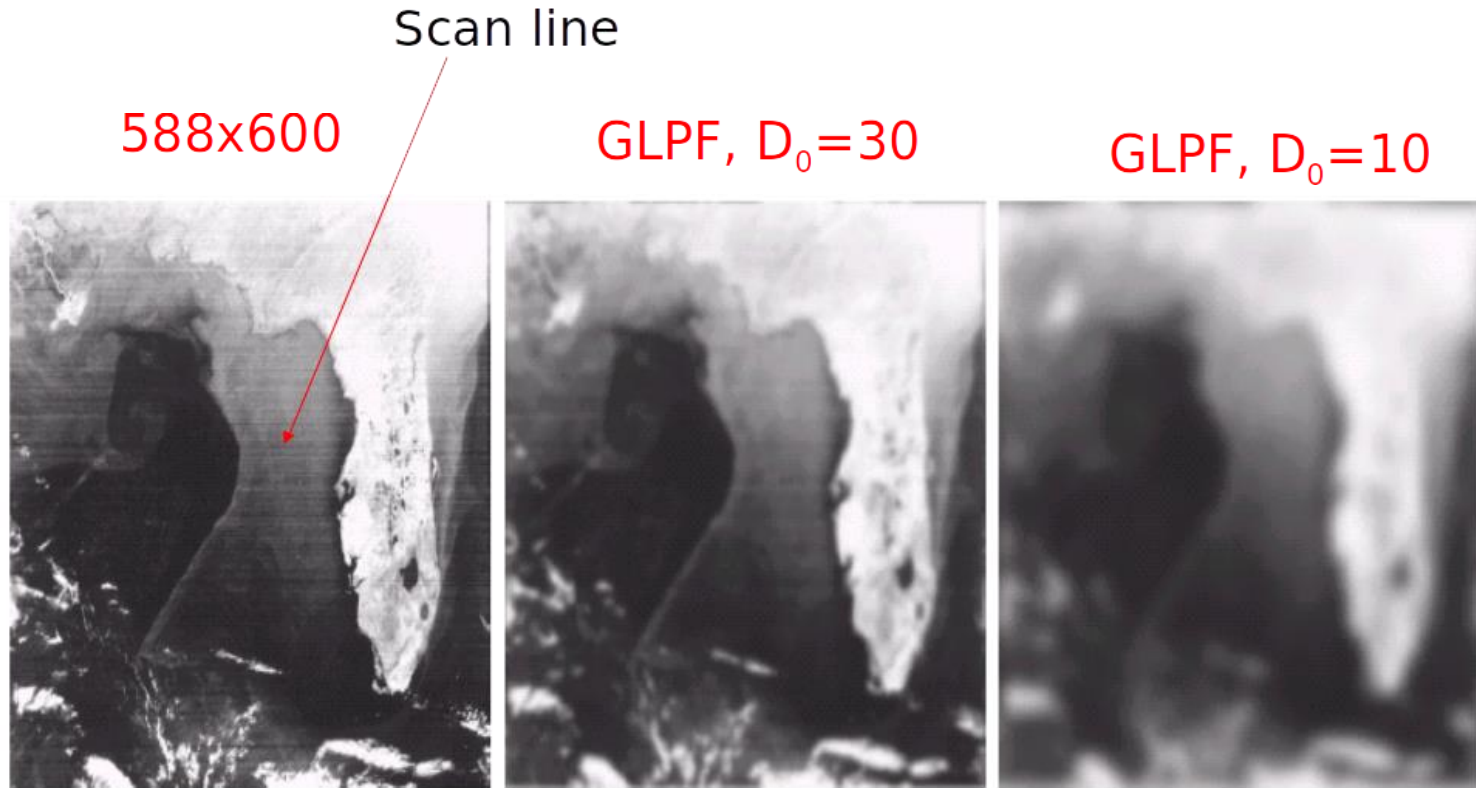


Original image

$D_0 = 100$

$D_0 = 80$

Practical Application 03



Sharpening frequency-domain filters

- Edges and fine detail in images are associated with high frequency components
- High pass filters – only pass the high frequencies, drop the low ones
- High pass frequencies are precisely the reverse of low pass filters, so:

$$H_{hp}(u,v)=1-H_{lp}(u,v)$$

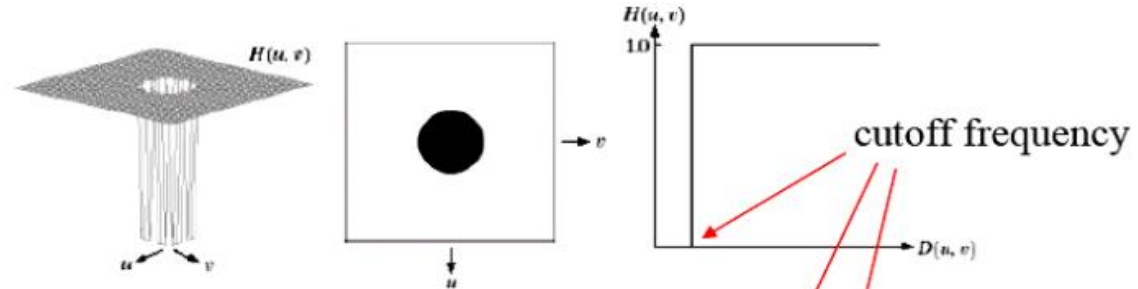
- Sharpening: high-pass filters

Sharpening frequency-domain filters

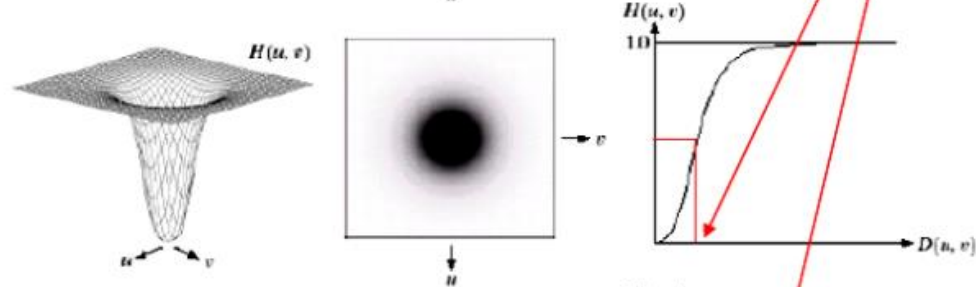
- Ideal high-pass filters
- Butterworth high-pass filter
- Gaussian high-pass filters
- Difference filters
 - Laplacian filters

Sharpening frequency-domain filters

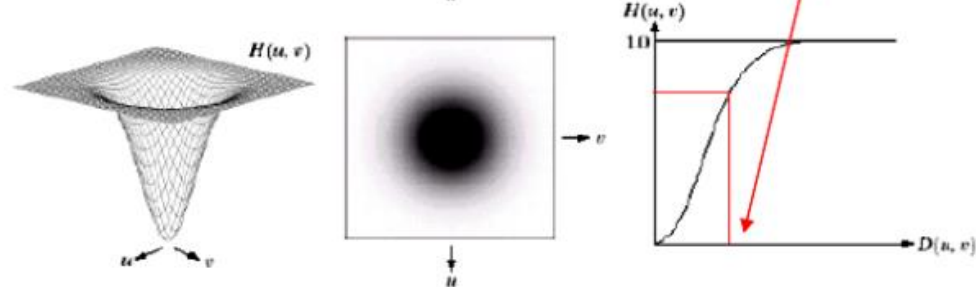
Ideal
highpass filter



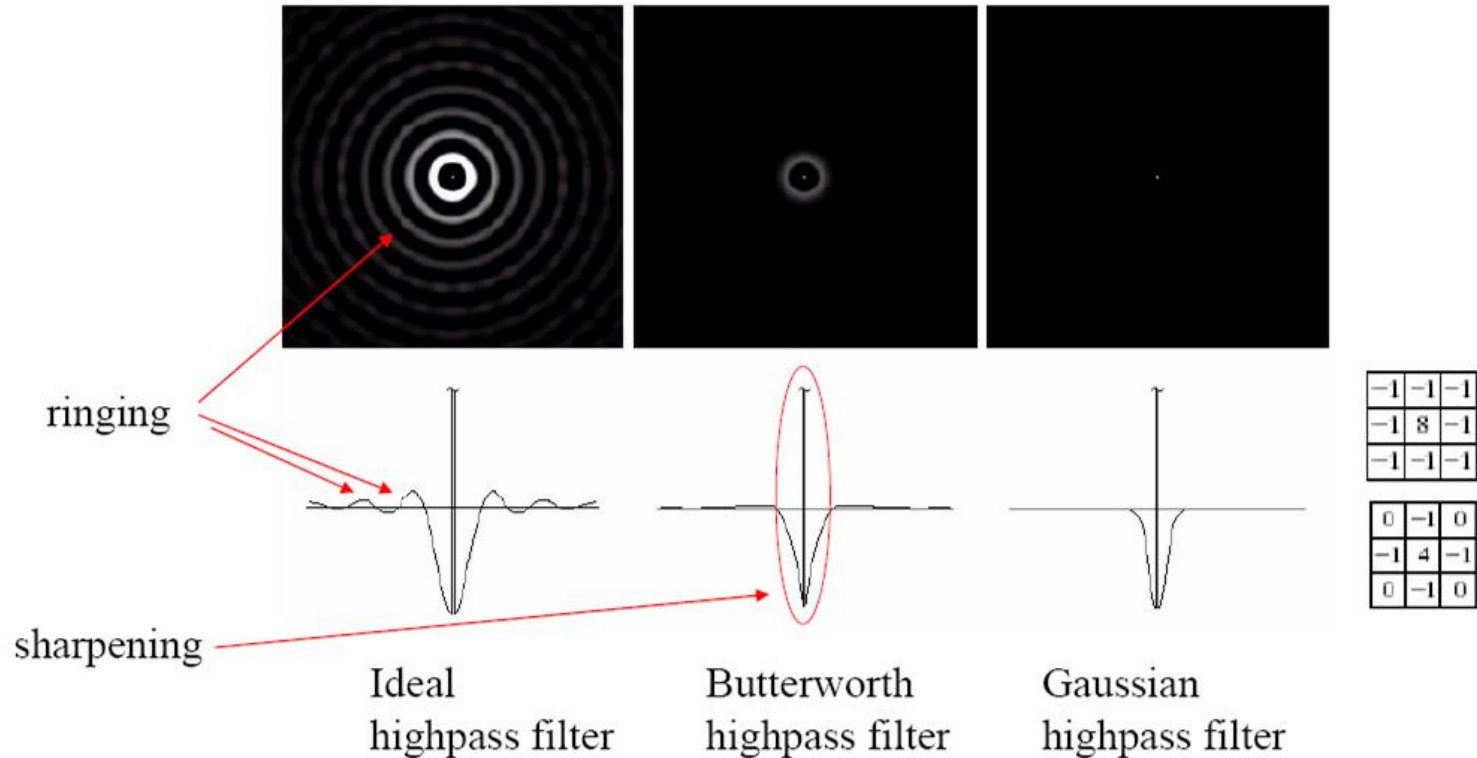
Butterworth
highpass filter



Gaussian
highpass filter

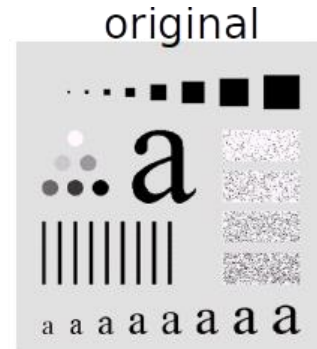


Spatial Representation of Highpass Filters



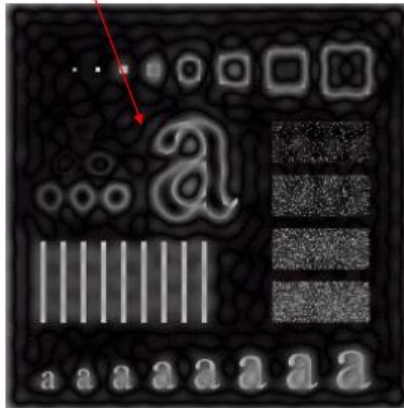
Ideal High-pass Filters

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) \geq D_0 \end{cases}$$

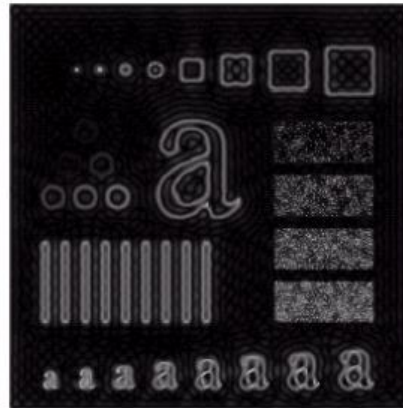


ringing

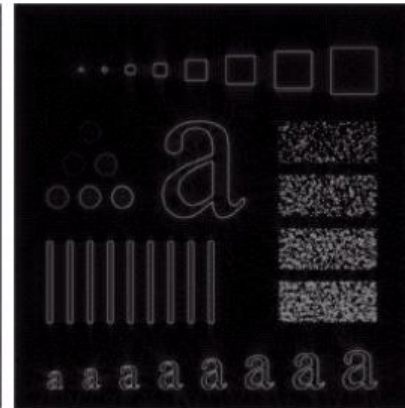
$D_0=15$



$D_0=30$



$D_0=80$



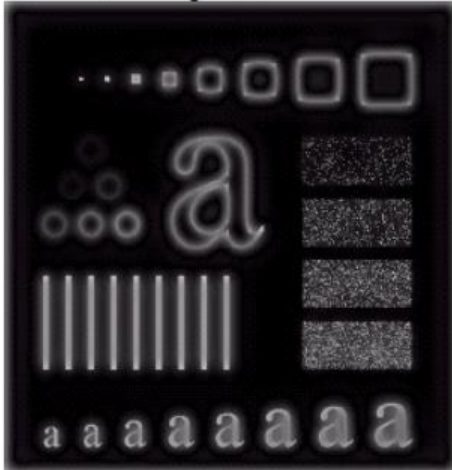
Sharpening & Ringing

Butterworth High-pass Filters

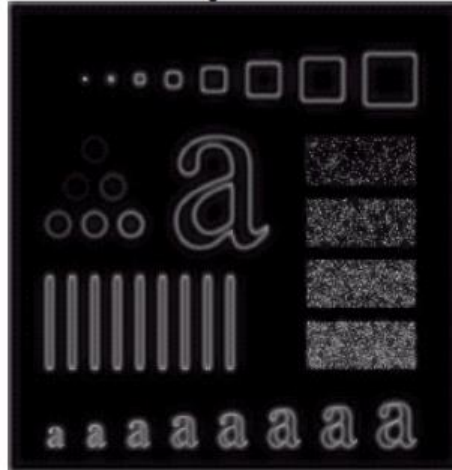
$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

Sharpening & No visible ringing

$n=2, D_0=15$



$D_0=30$



$D_0=80$

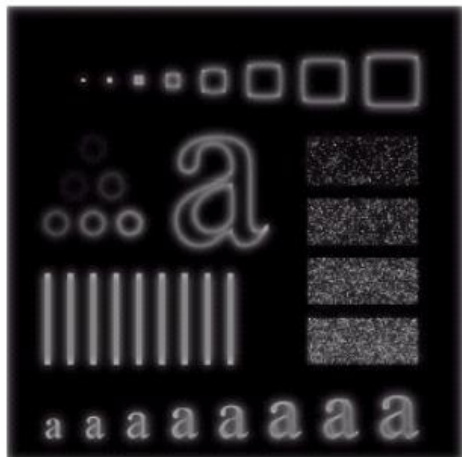


Gaussian High-pass Filters

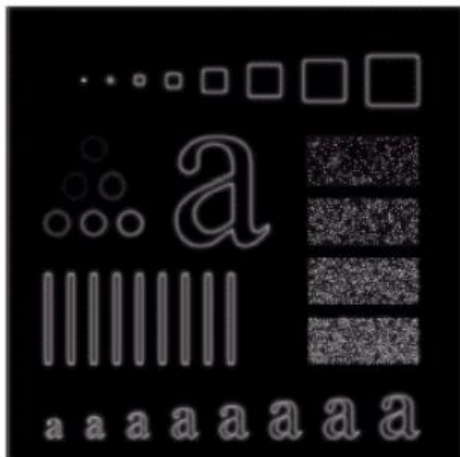
$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

Sharpening & No ringing

$D_0=15$



$D_0=30$



$D_0=80$

